

Blue Apron Training

C-Block / Learning Computer Automated Machining (CAM)

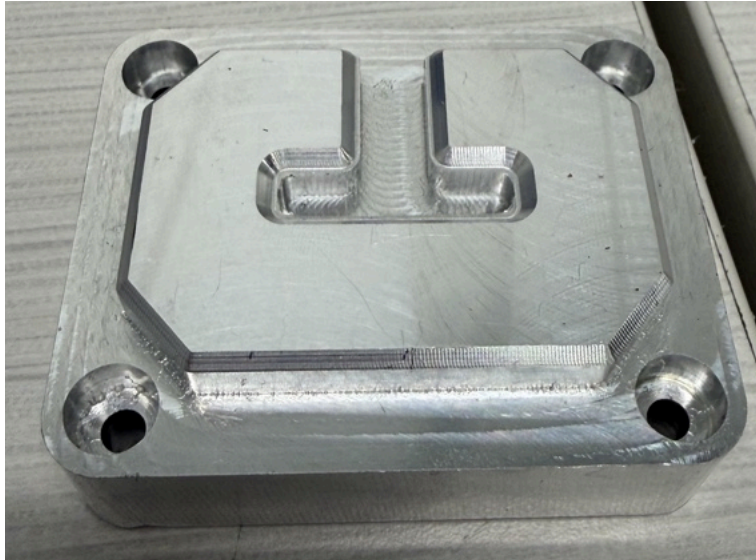


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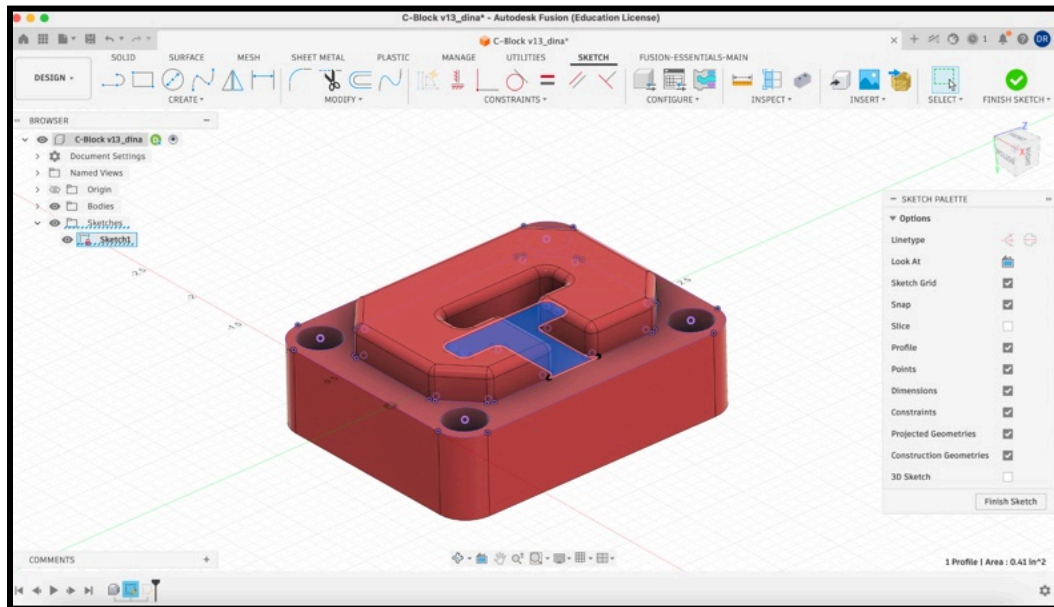
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Part 0: Make Preliminary Sketches

Before creating the CAM, we create a few sketches that will be referenced by later operations.

Sketch 1: Inner C Cut

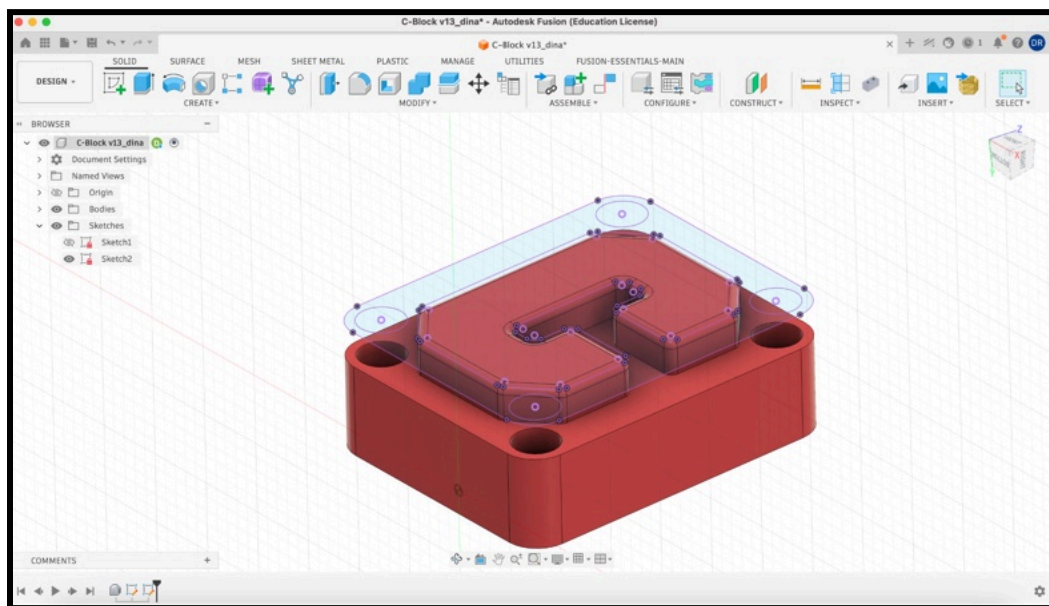
1. Open the C-Block design in Fusion 360 and import the C-Block STL file.
2. Create a sketch on the top surface of the base



3. Use the Project feature to project the relevant model edges into the sketch
4. Connect the projected geometry with lines as necessary to create a closed profile

Sketch 2: Outer C Outline

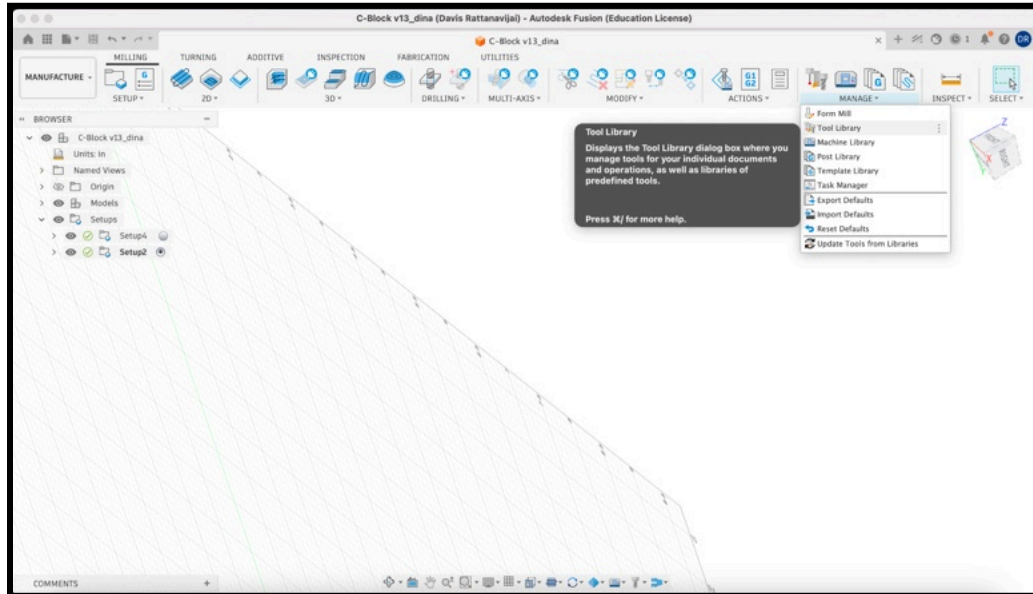
1. Similarly, create another sketch on the top surface and project the C-block outer edges



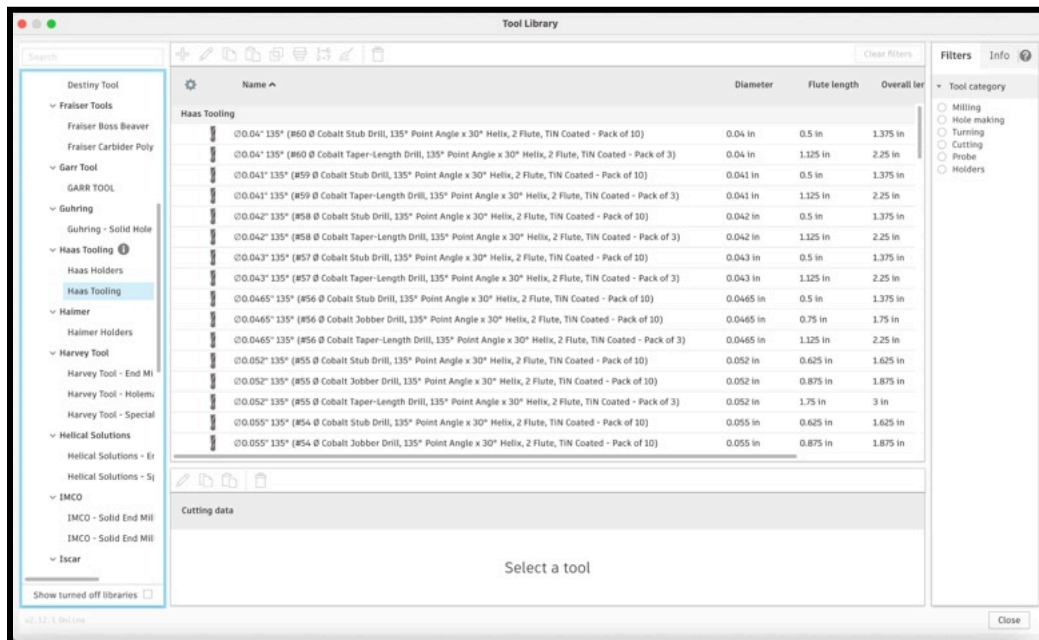
Part 1: Set Up the Tool Library

We first need to add the tools that will be used to manufacture the C-Block.

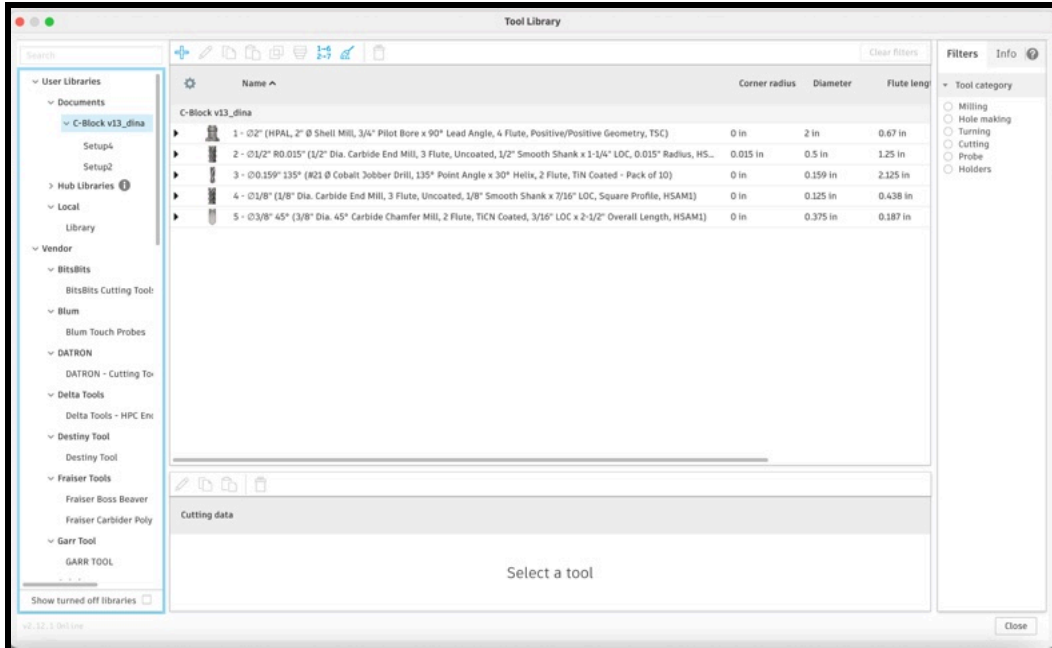
1. In Fusion 360, use the dropdown in the upper-left corner to switch from Design to Manufacture
2. Navigate to Manage → Tool Library.



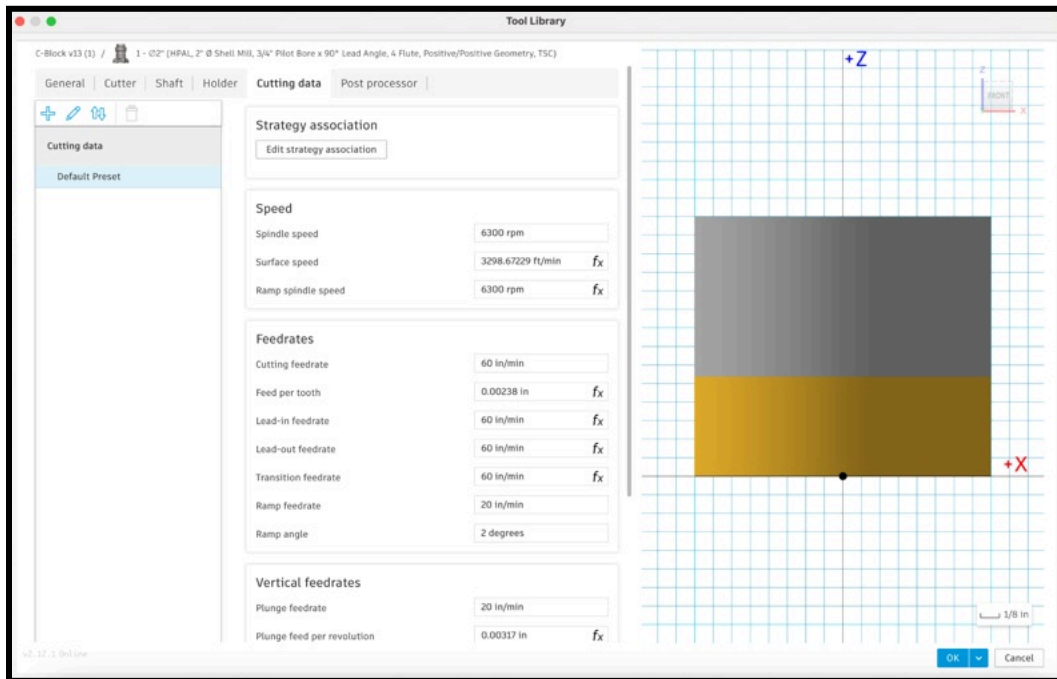
3. Select the HAAS Tooling library.
4. Use the filters on the right side of the Tool Library to locate the required tools.



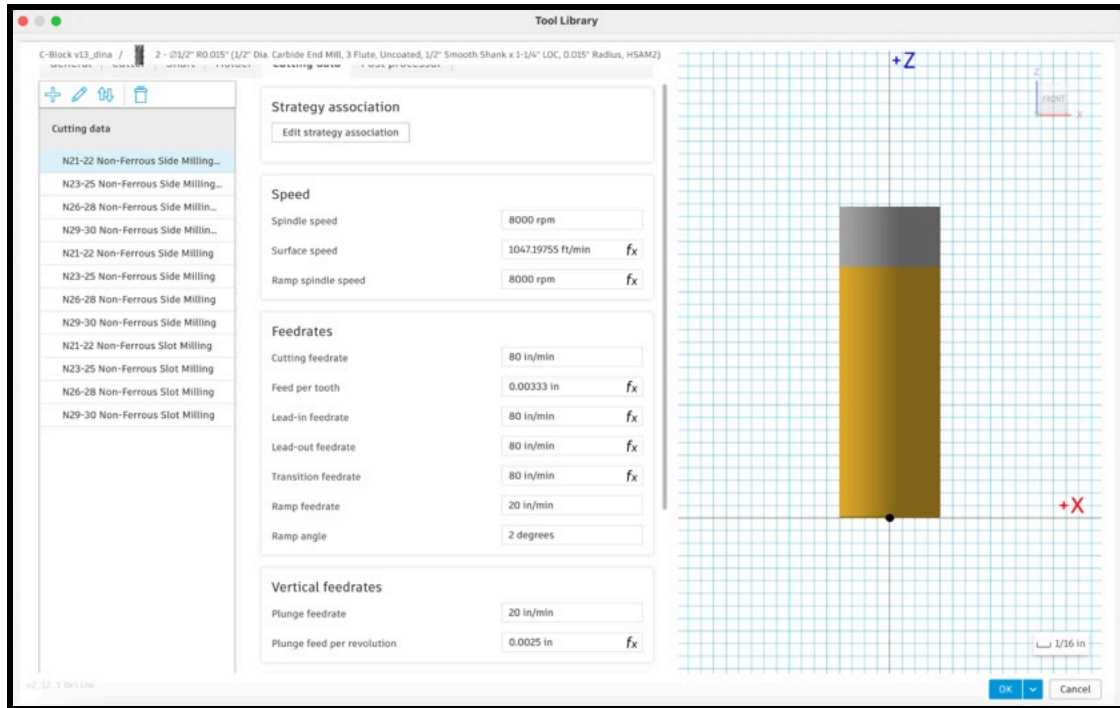
5. Find and add the following five tools needed to machine the C-Block.



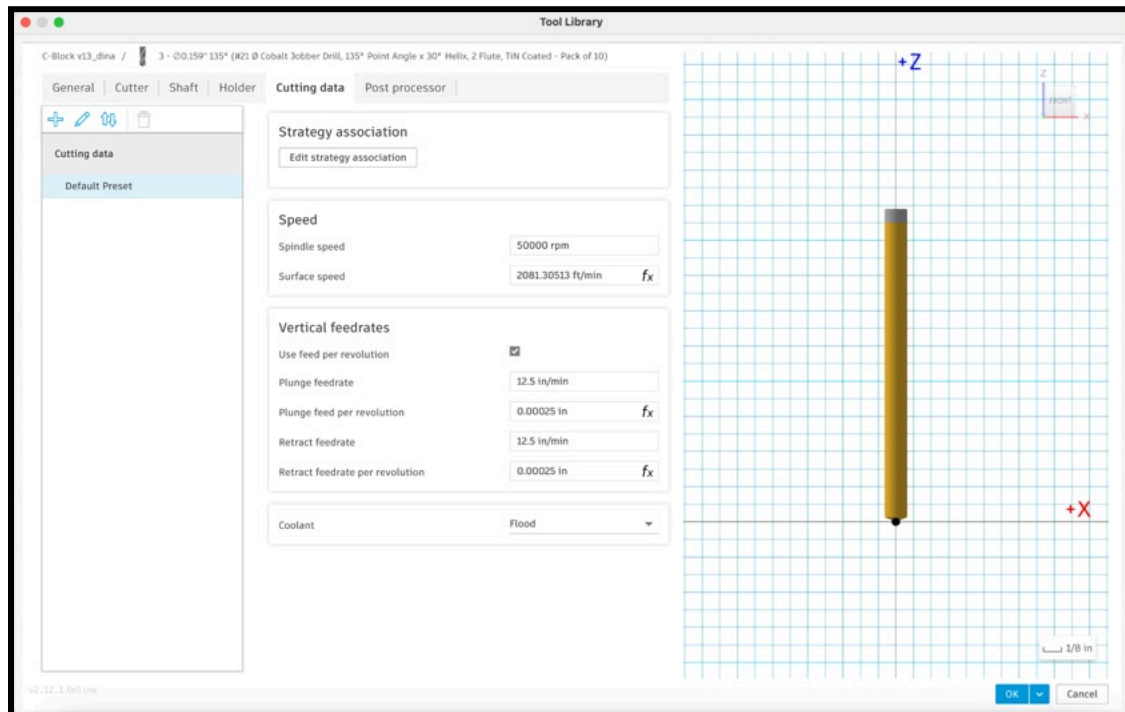
6. Once you have found each tool, double-click it and edit the Cutting Data to match the desired machining parameters. Below are the recommended parameters for each tool.



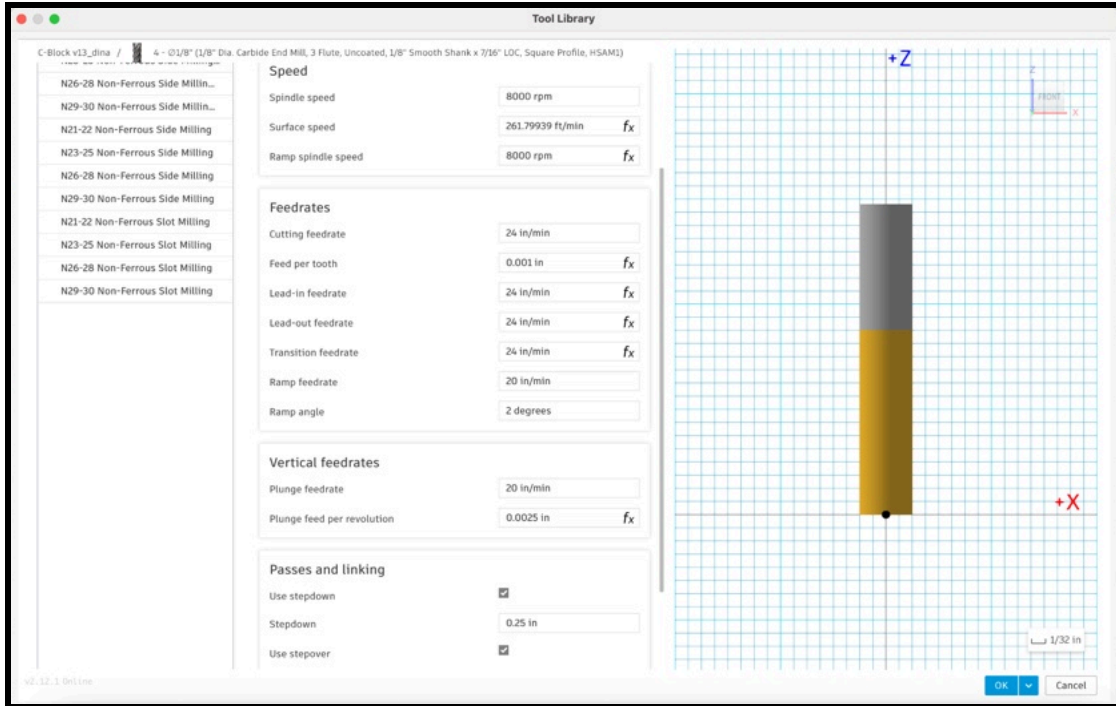
Shell Mill



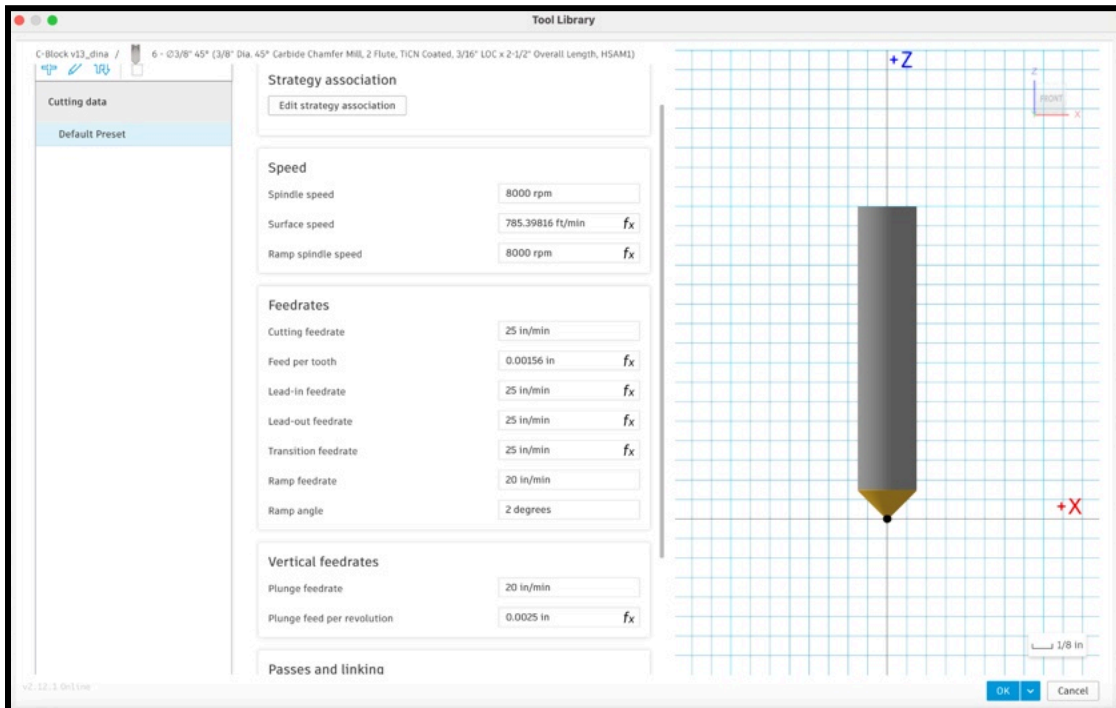
0.5" End Mill



0.159" Drill Bit



1/8" End Mill

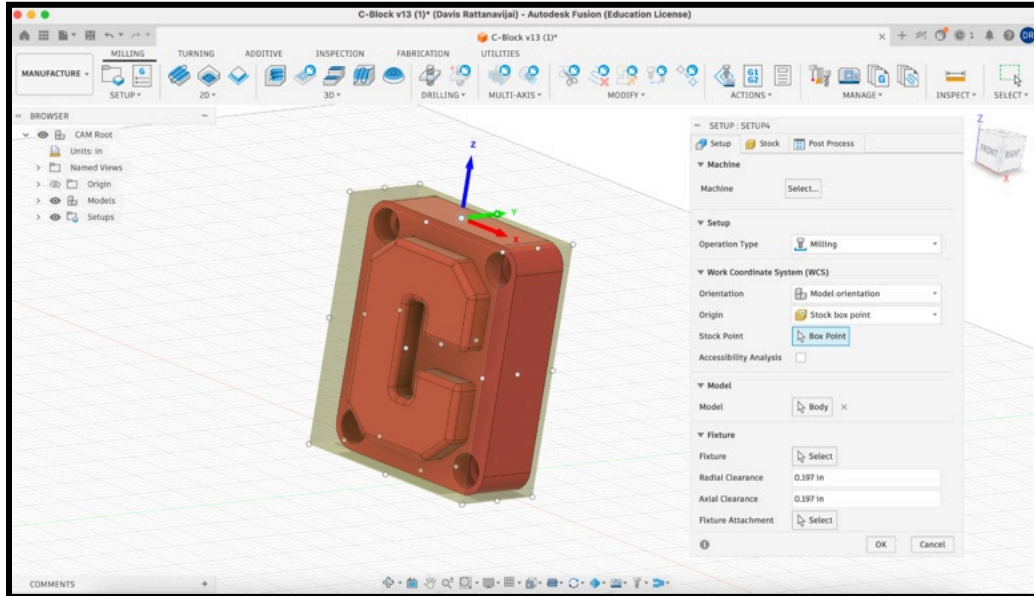


Chamfer Bit

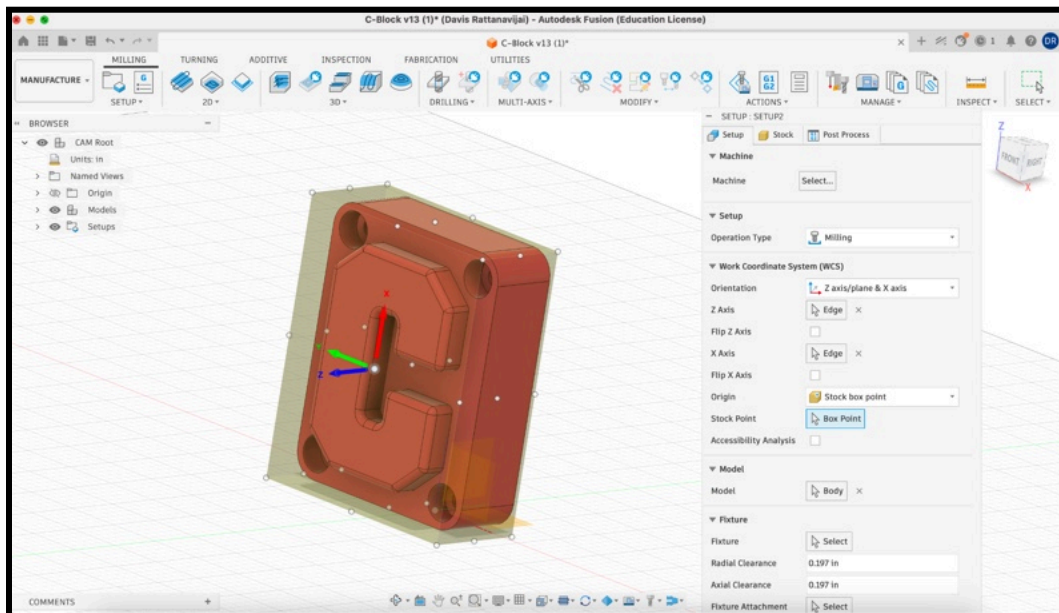
Part 2: Set Up Coordinates and Stock

Before generating toolpaths, we establish the coordinate system and define the raw stock.

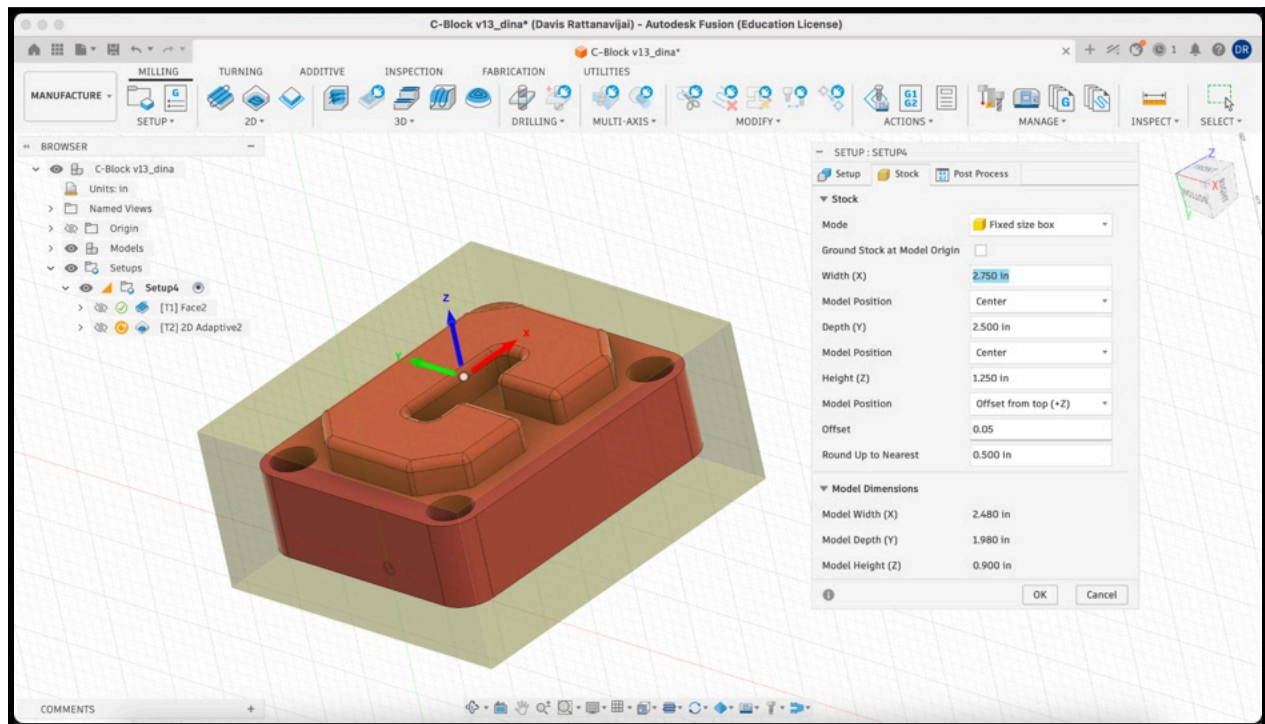
1. Create a new Setup from the Setup tab
2. Set the Operation Type to Milling



3. Set the Orientation to Z Axis/Plane & X Axis. This allows you to select the X- and Z-axes, while the Y-axis is automatically calculated using the Right-Hand Rule (RHR).
4. Select the appropriate geometry to define the axes. Select the appropriate lines to define each axis, and use the Flip Axis checkboxes if needed. The Z-axis should point upward, and the X-axis should run along the long side of the block.
5. Set Origin to Stock Box Point.
6. Select the center of the top face of the stock as the origin. Later, we will probe this point.



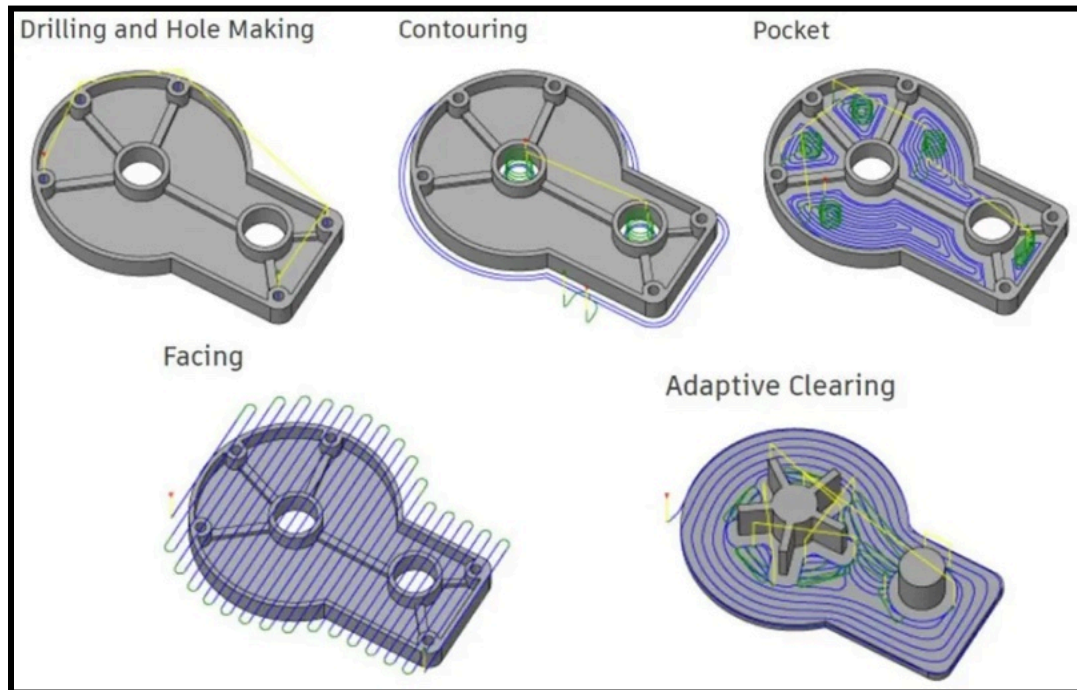
7. Go to the Stock tab
8. Set Mode to Fixed Size Box
9. Enter the measured X, Y, and Z dimensions of the aluminum stock
10. Set the Stock Offset to 0.050 in



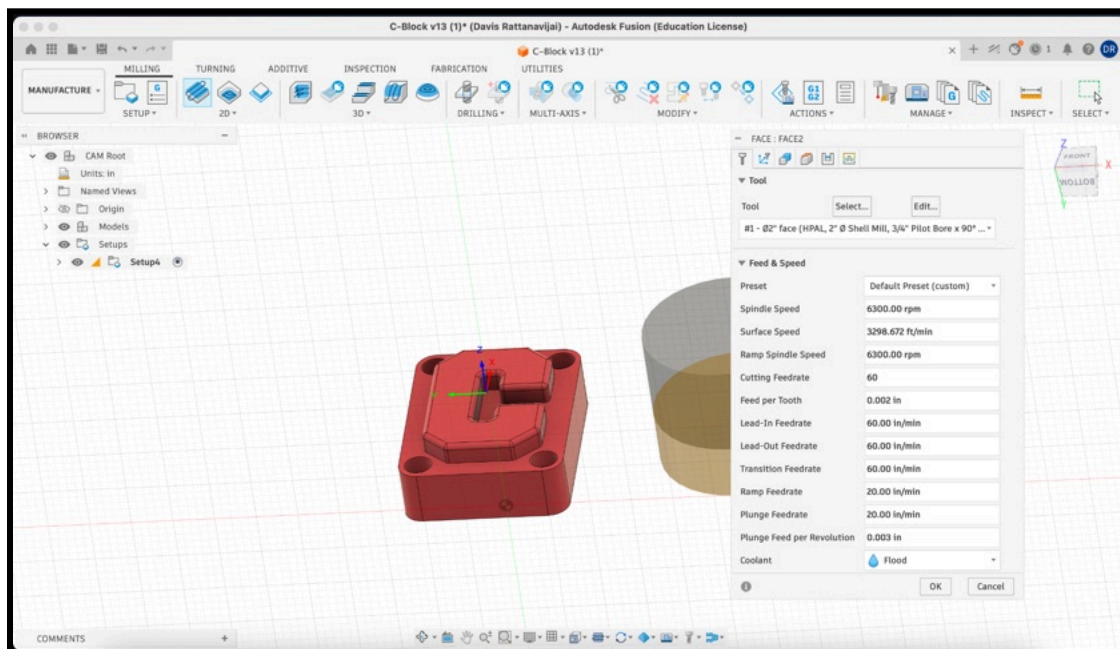
11. Click OK. The yellow opaque rectangle represents your stock.

Part 3: Face the Top Surface

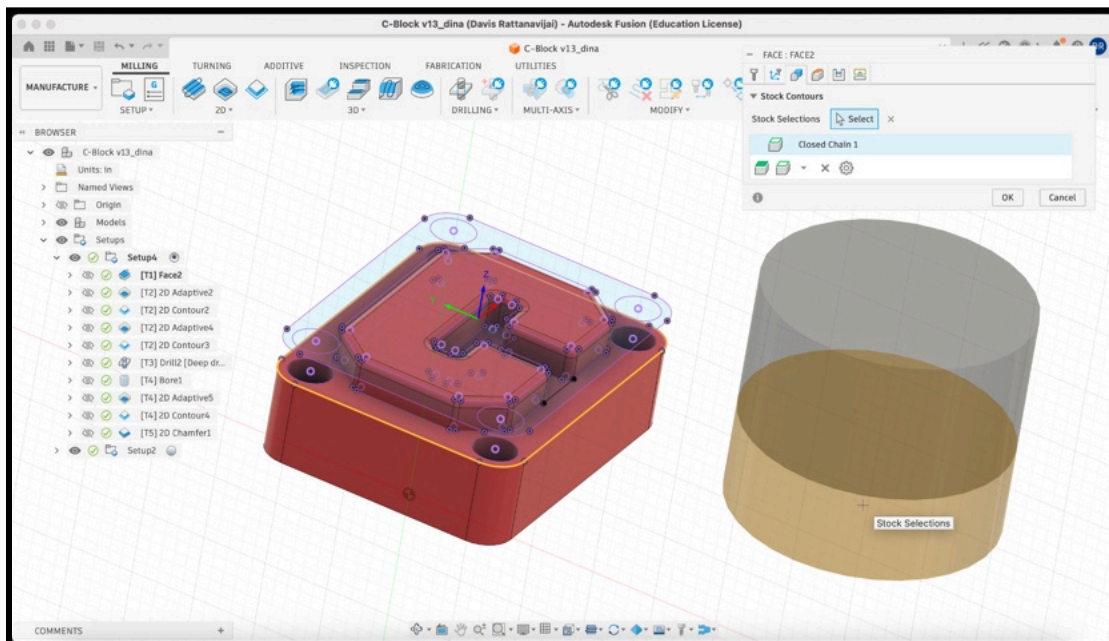
Before we begin, let's go over the five main 2D operations in Fusion 360 which are Drilling, Contour, Pocket, Face, and Adaptive Clearing. In the image below, the blue lines represent where the tool is actively cutting material, while the yellow lines represent the tool's movement when it is not cutting.



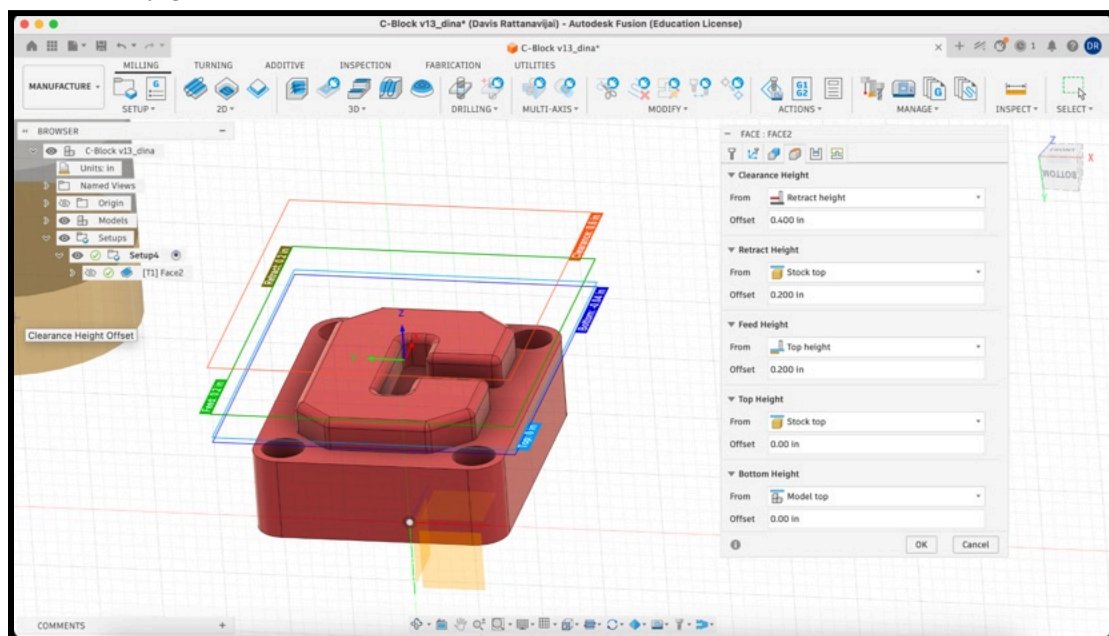
1. To begin to face the top surface, select 2D → Face
2. Select the shell mill and verify the tool diameter, feeds, and speeds.



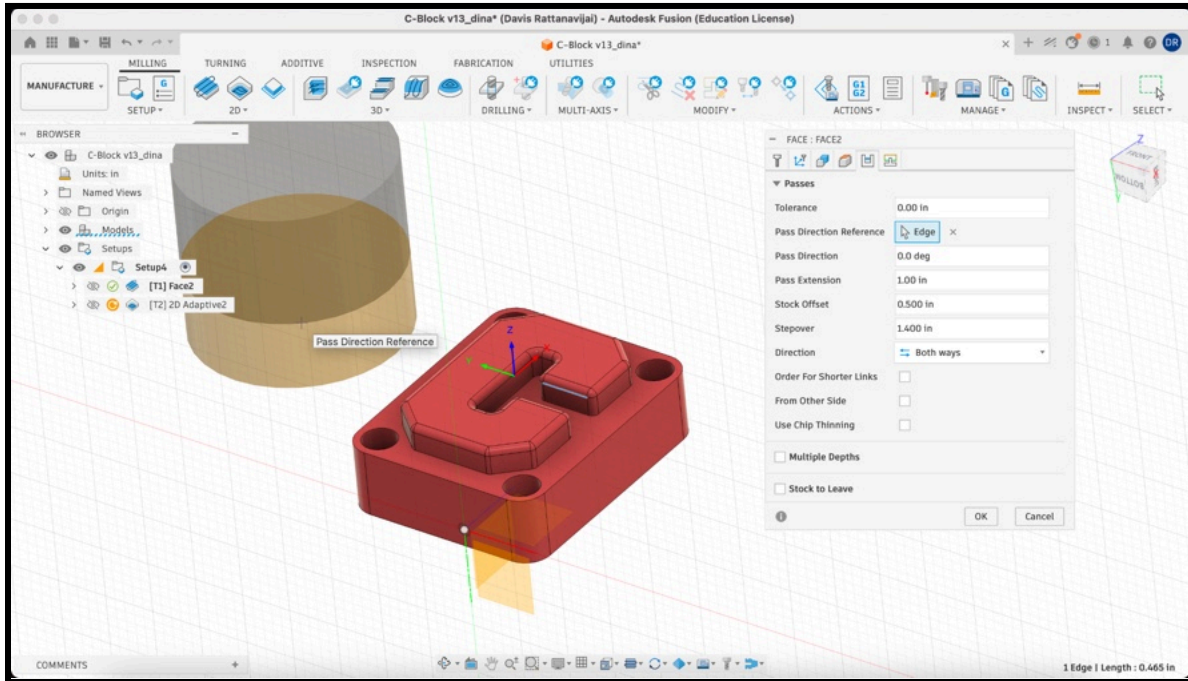
- Go to the Geometry tab. We can select either a line or face, but we want to choose one that encapsulates the entire area of the part. In this case we chose a closed chain that uses the perimeter of the C-Block. We could have also selected the face of the bottom block.



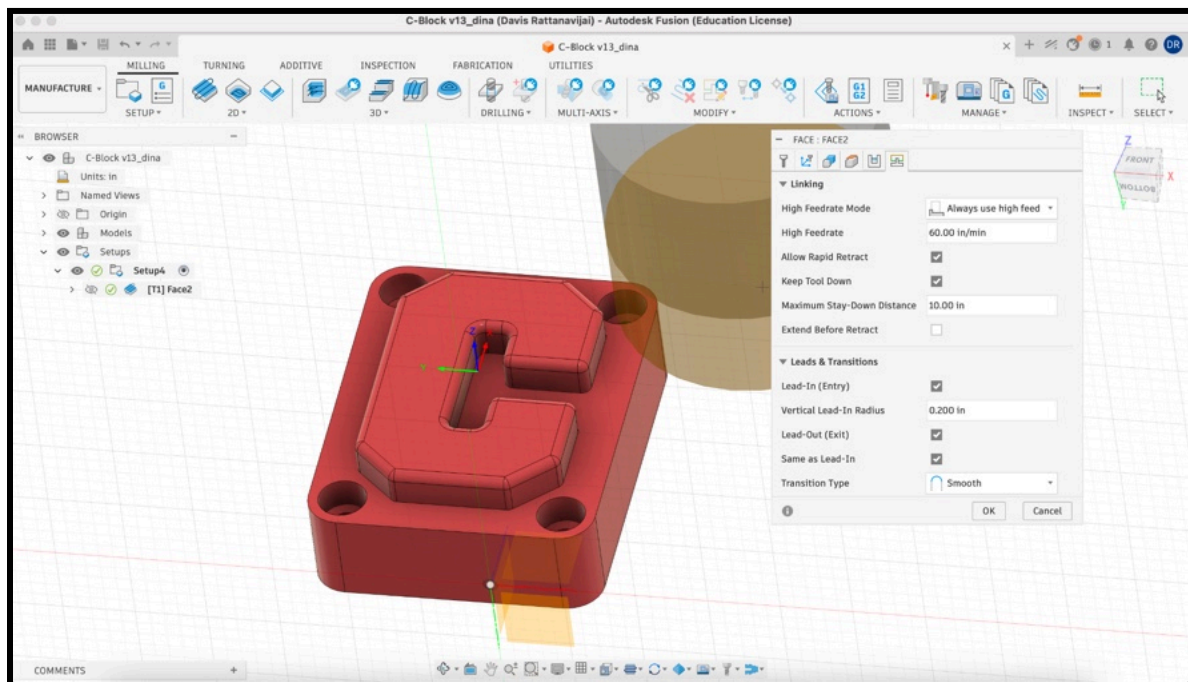
- Then to the Heights tab, which establishes the important heights for the tool to reference. The clearance height should be well above the stock (not just the C-block CAD). Clearance Height is the highest plane the tool rapids to when moving between operations, and must clear the stock and fixtures, not just the model geometry. Retract Height is where the tool rapids to between passes within an operation, set above stock top. Feed Height is where the tool switches from rapid to feed rate before engaging material. Top Height marks the top surface of the stock/model where cutting begins, and Bottom Height is the deepest point the tool will cut to. The preset parameters are usually good



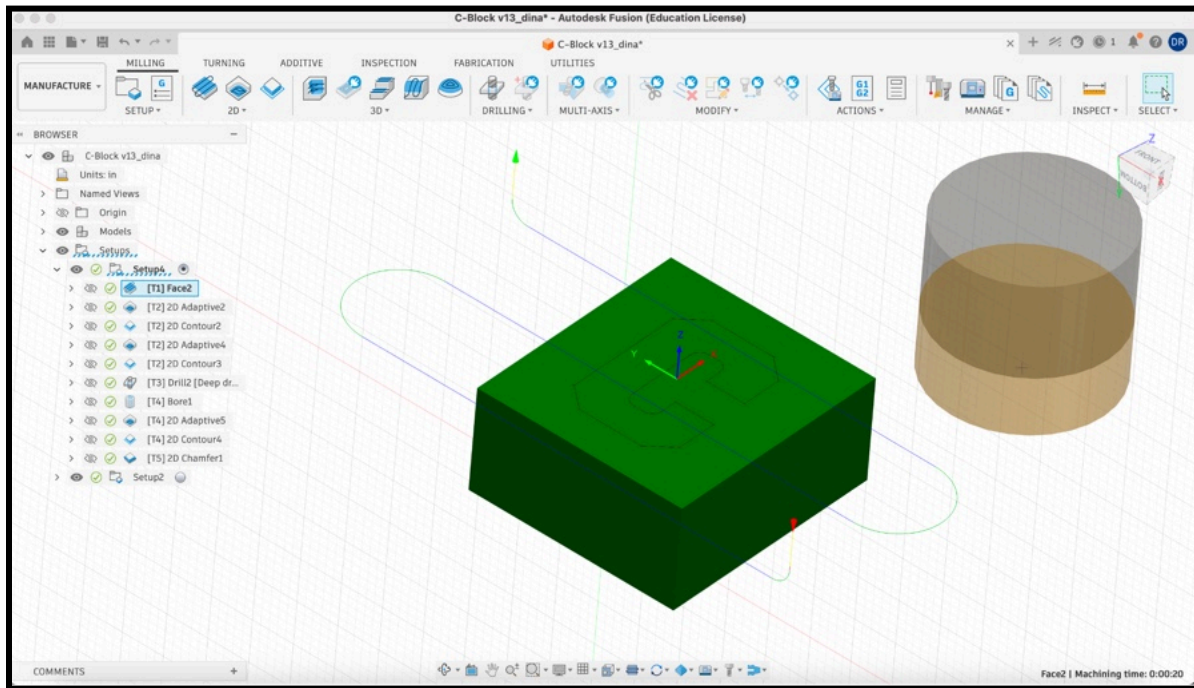
- Then go to the Face tab and select the pass direction reference by clicking on a line in the direction you want the cutter to move. Generally, you want the cutter to move in the direction the vise is clamping to minimize slippage of the stock. You should also set the stock offset to be large enough to ensure that it cuts the entire piece. Once you press OK at the end, you can view the toolpath. Make sure it faces the entire surface.



- Go back to the Linking tab and double-check the values. They should look like the ones shown below.



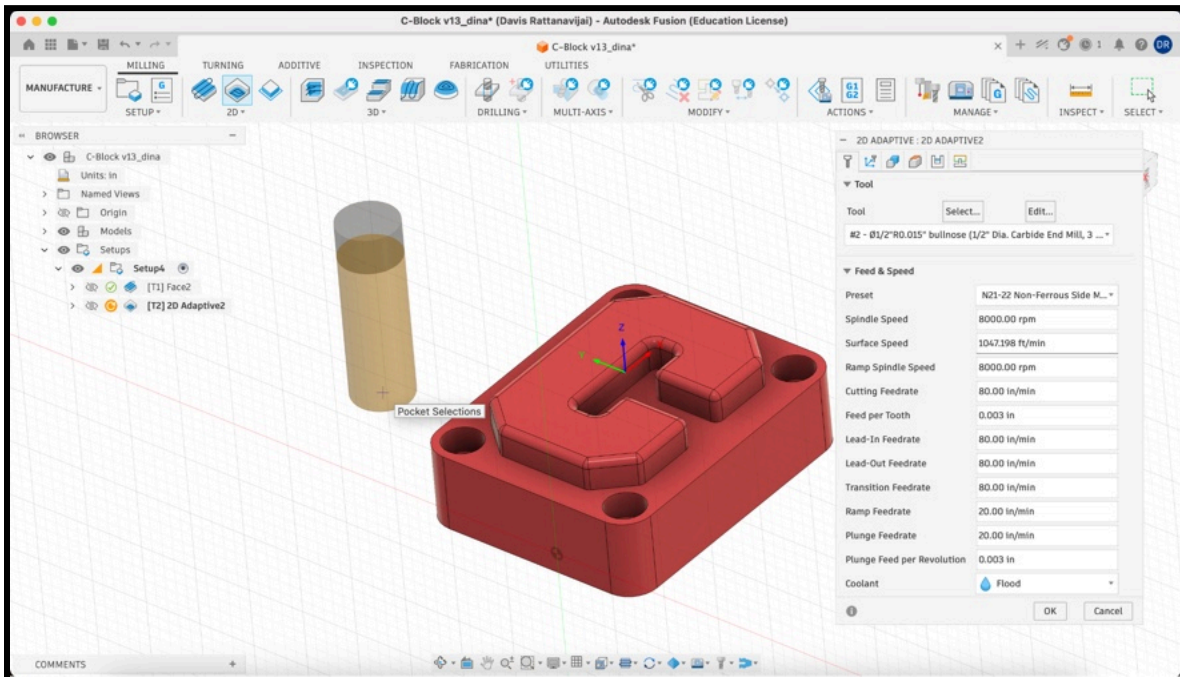
- Once you press okay, it will show you what the cut looks like. The green is what is left of the stock. Verify that the whole surface was faced up until the C cutout like shown.



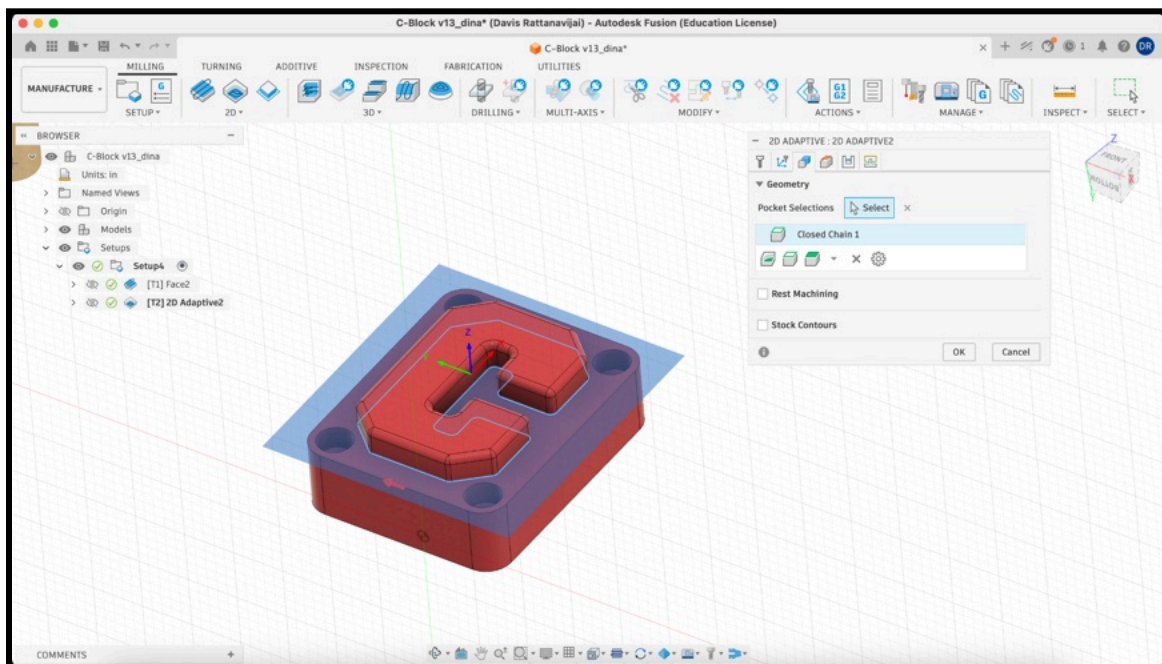
Part 4: Milling the C Outer Outline

In this part we will mill out the general outer profile of the C.

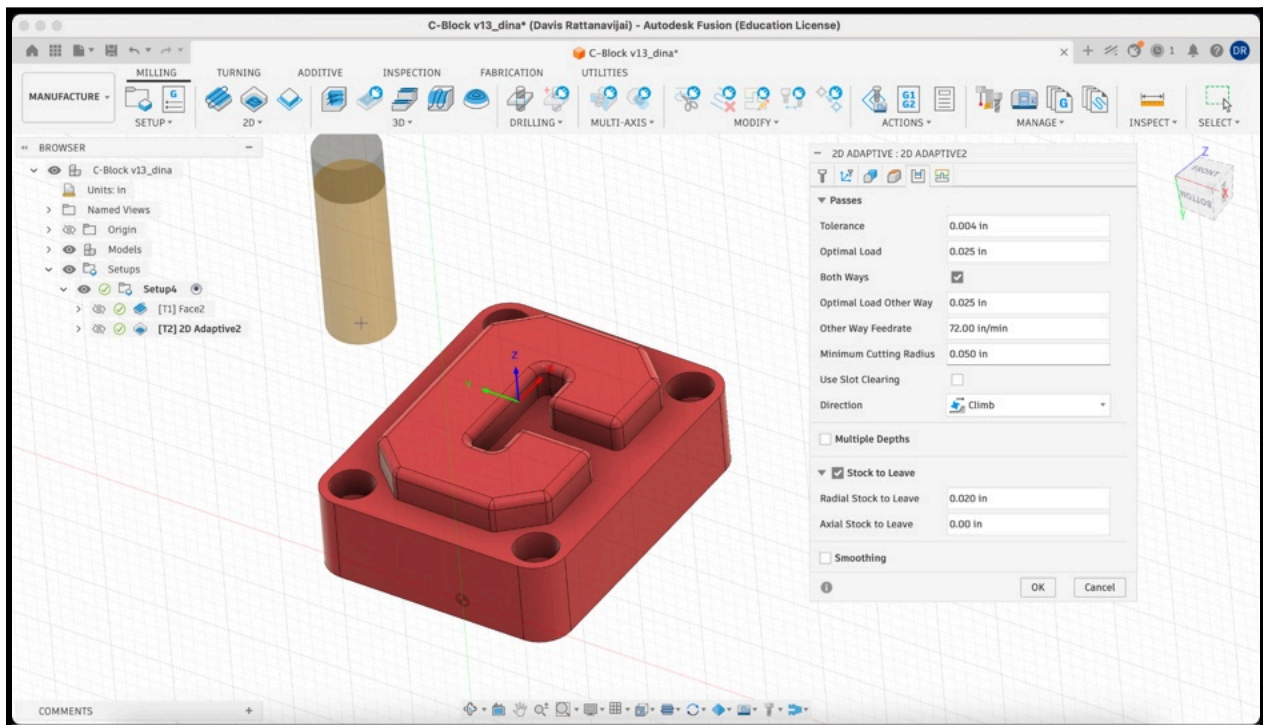
1. Select 2D Adaptive. This is a high-speed roughing strategy that quickly removes bulk material while maintaining a consistent tool load to reduce the risk of tool breakage.
2. Select your 1/2 inch end mill. Verify that the feeds and speed is what you set in the beginning and that they are reasonable.



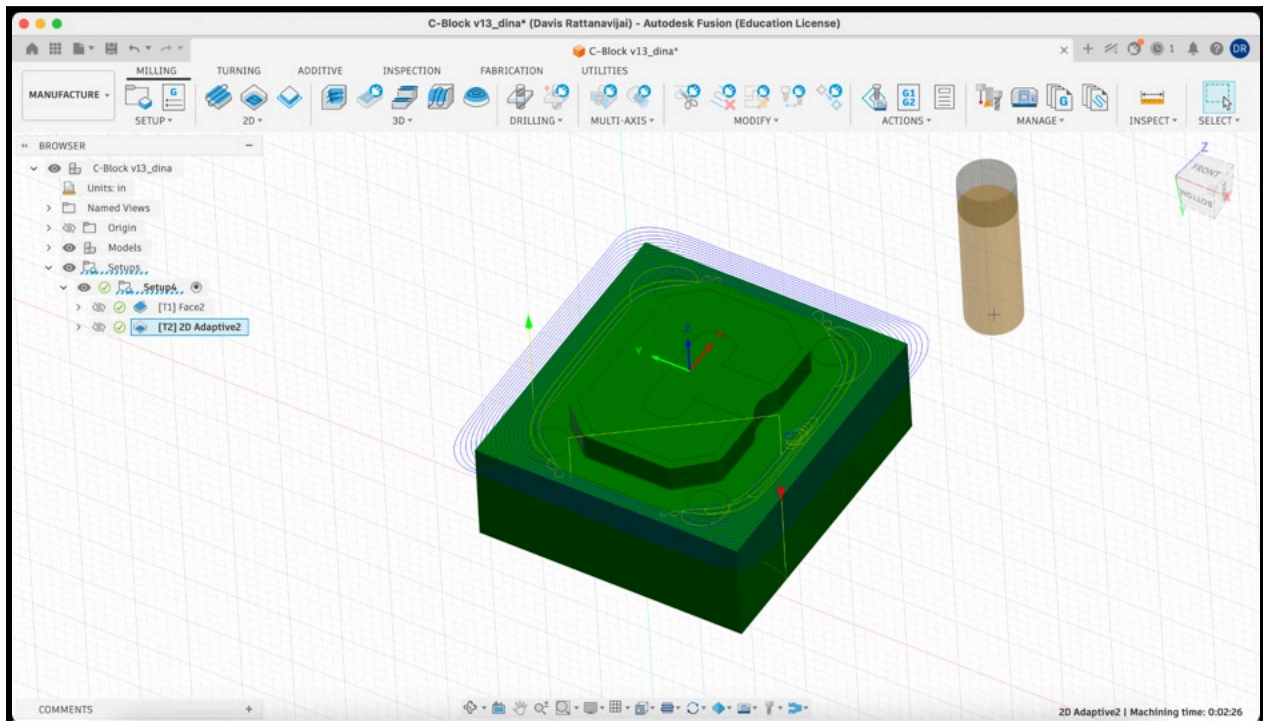
3. Next, move to the geometry tab and select the closed chain option and then select the C outline.



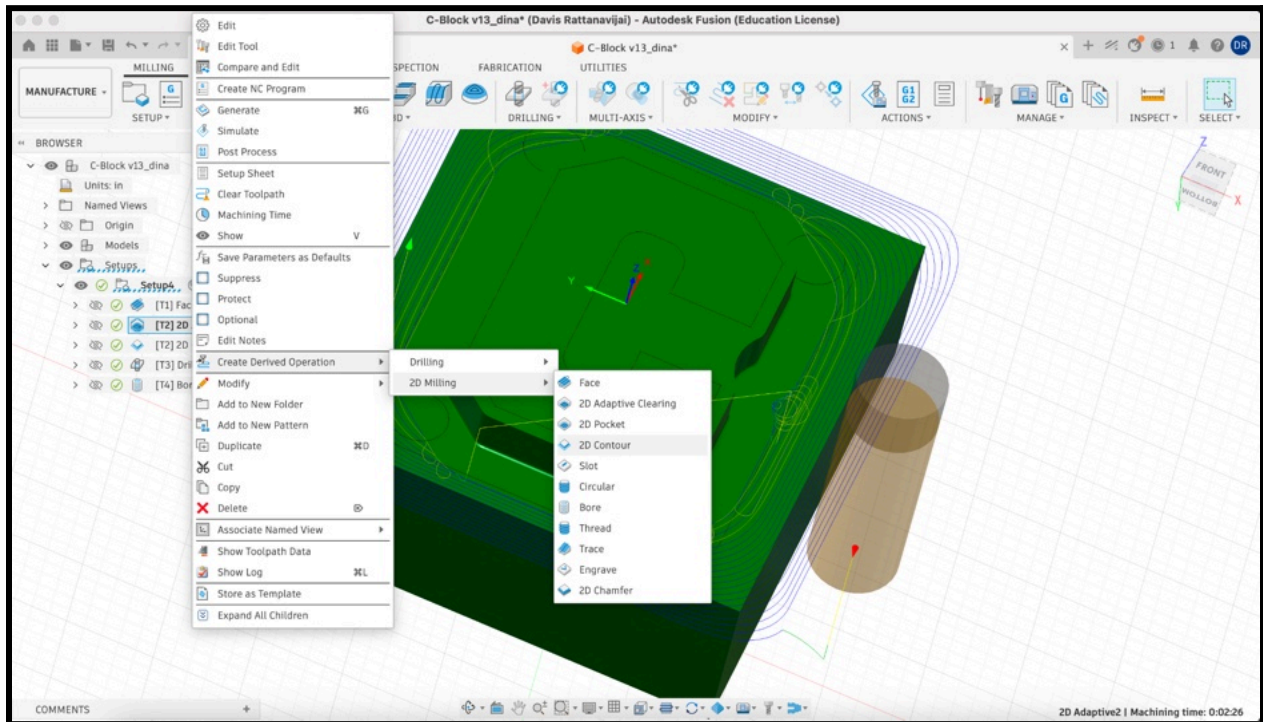
4. In the Passes tab, set the Optimal Load to the amount of material you want to remove. 25 thou is reasonable. Check both ways. Set Axial Stock to Leave to 0 so the end mill fully faces the top surface. Set Radial Stock to Leave to around 0.02 for the finishing Contour operation that is next.



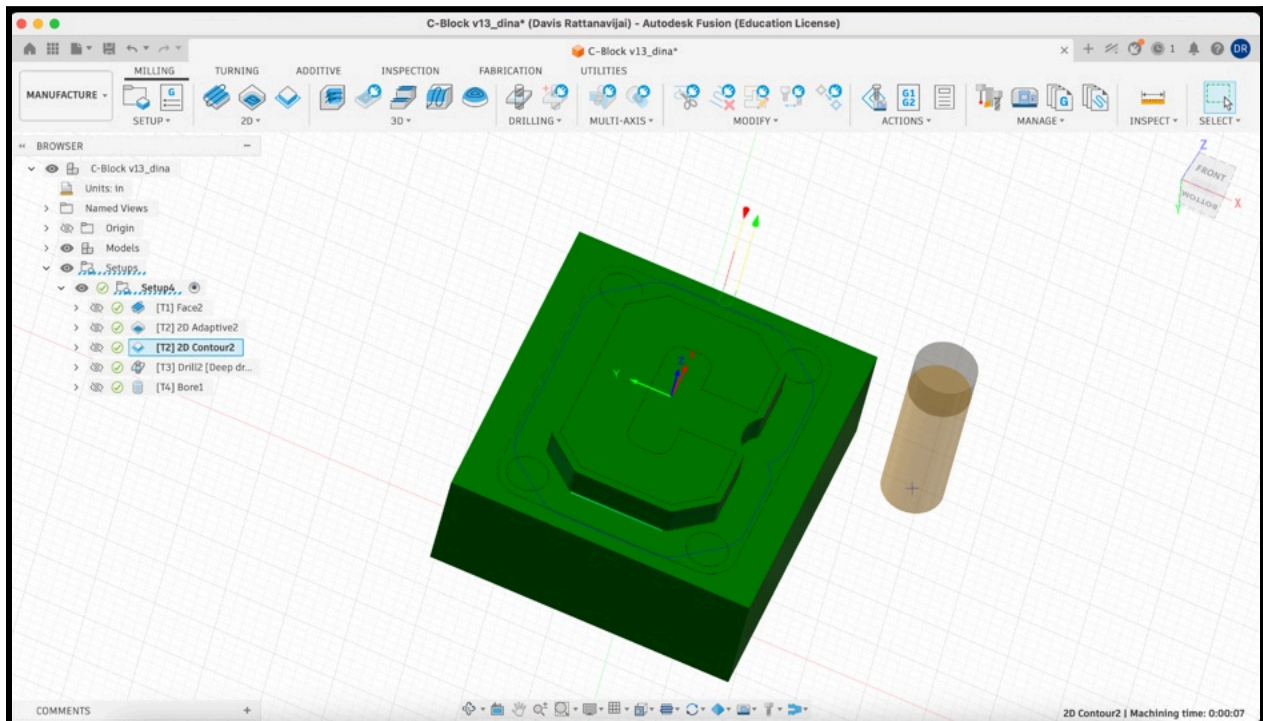
5. Once you click OK, view the toolpath and make sure you see the general C outline being cut out. Check that the passes look reasonable and that the tool is removing a reasonable amount of material with each pass.



6. Next, we will create another milling event to remove the 20 thou of stock left from the adaptive operation. We will use the same end mill and select the Contour option. Instead of entering all the settings again, we can create a Derived Operation, which automatically carries over the existing values. We will create a derived operation for the 2D Contour event.



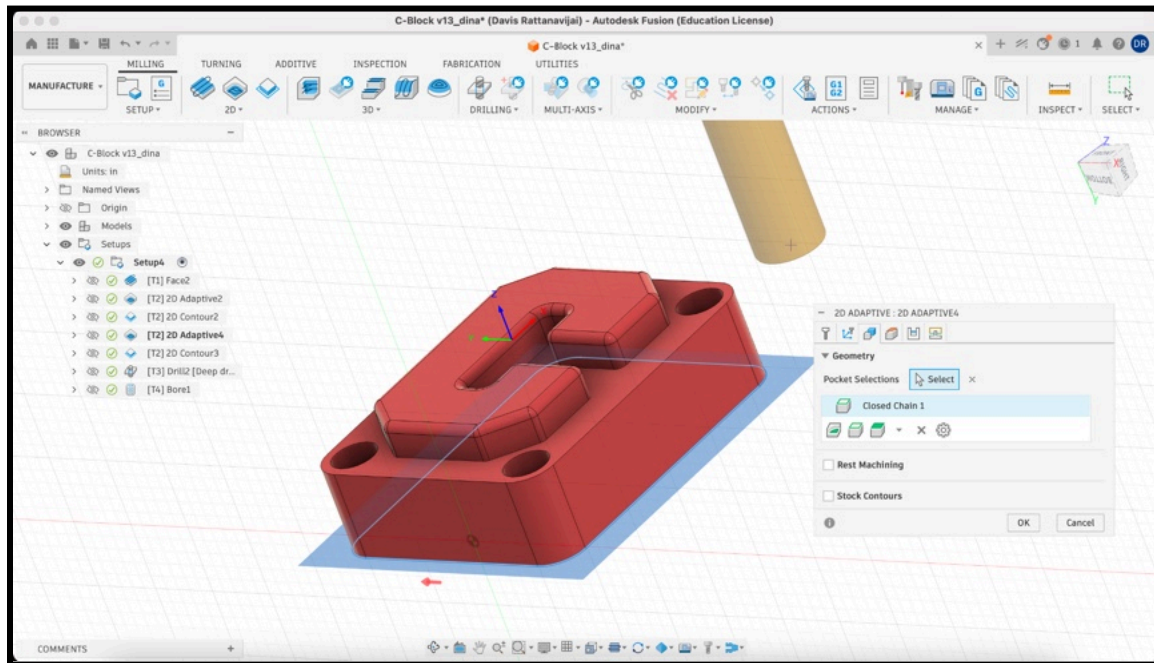
7. You can keep all the previous values, except make sure Radial Stock to Leave is set to 0.00. Your toolpath should look like this.



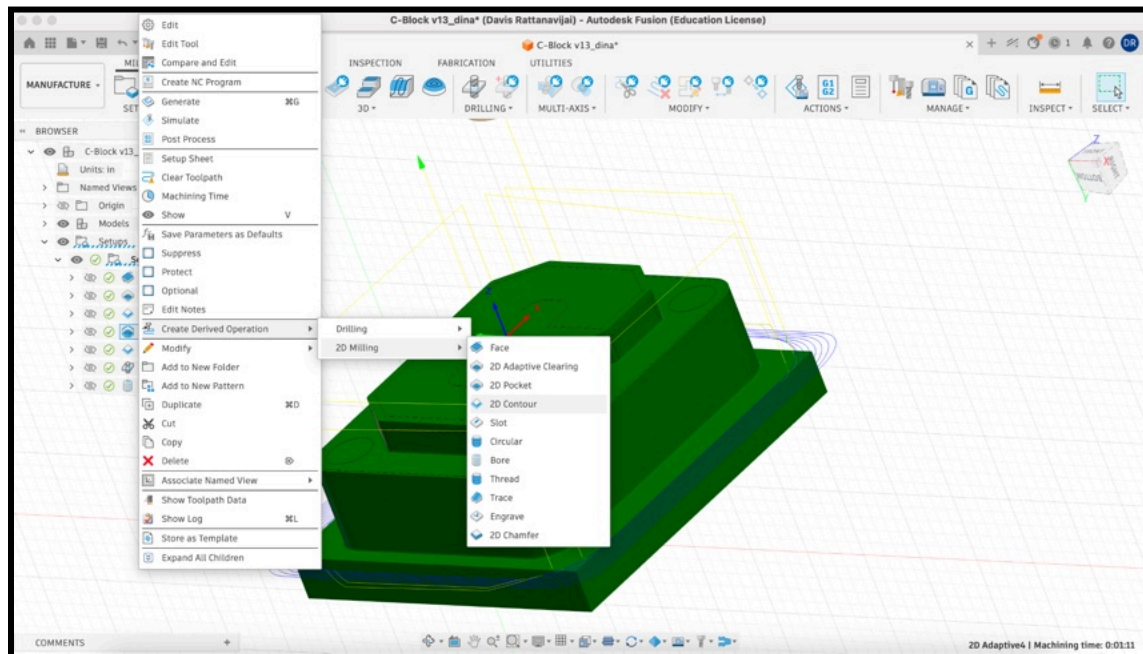
Part 5: Milling the Block Outline

In this part, we will mill off the sides of the block.

1. Select the 2D Adaptive event, but continue using the end mill. Select the Closed Chain at the bottom of the face. The settings should be mostly the same, but be sure to set Radial Stock to Leave to 0.02, as we will still use a contour event.



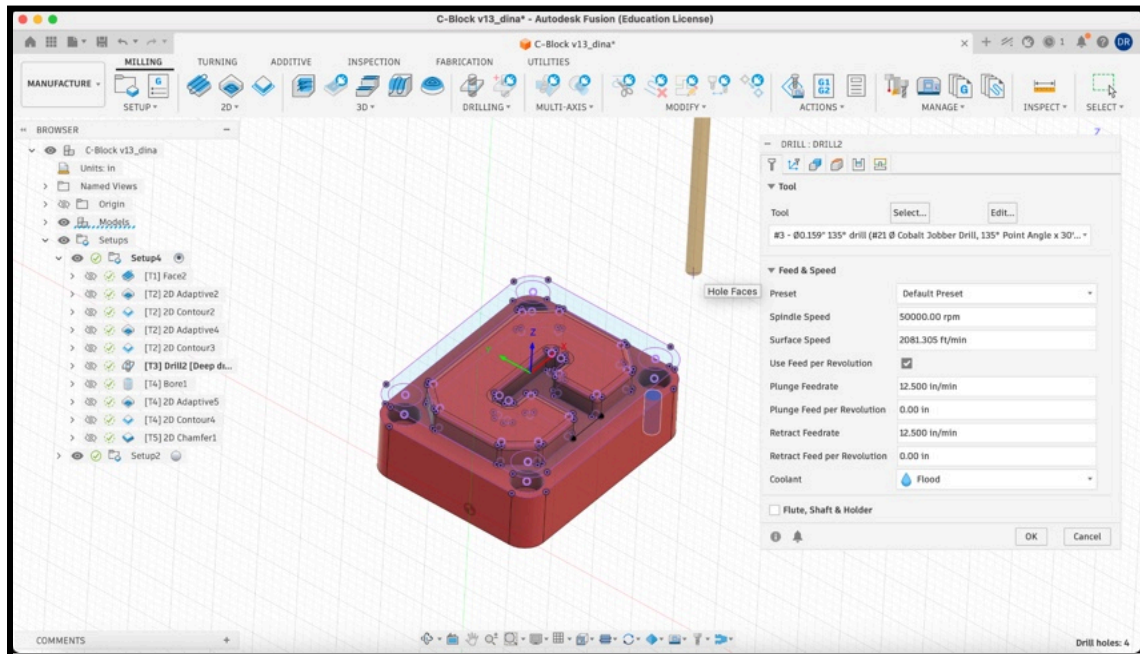
2. Next, use a Contour event derived from your 2D Adaptive event. After that, verify that your toolpath looks reasonable.



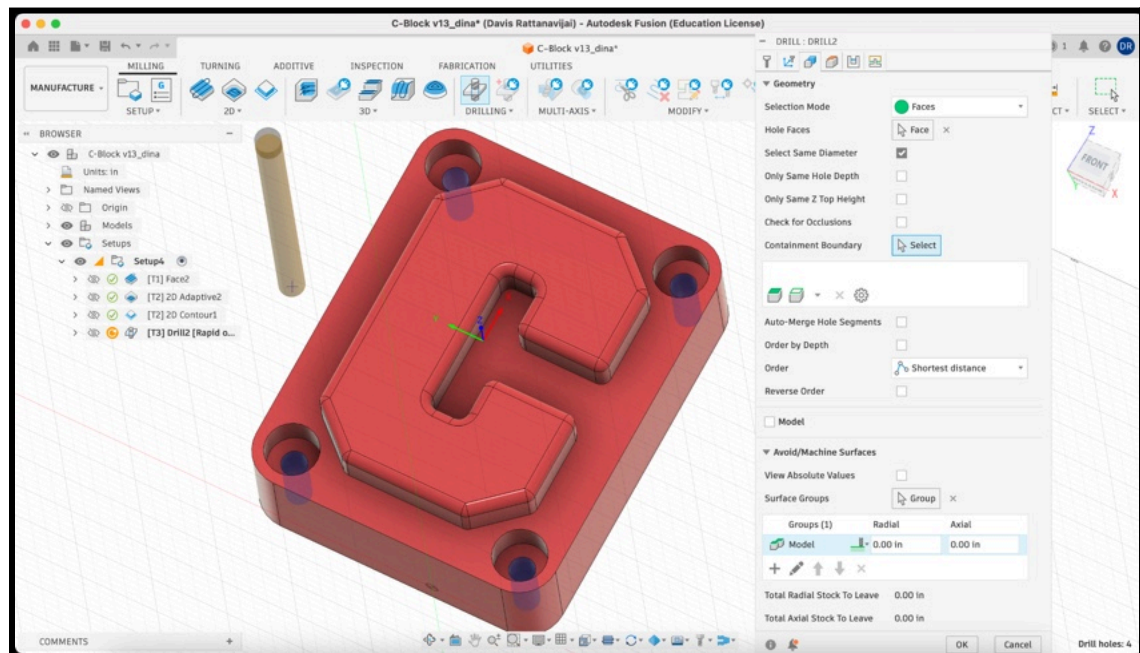
Part 6: Drilling the Holes

Now we are going to drill the four holes in the corners.

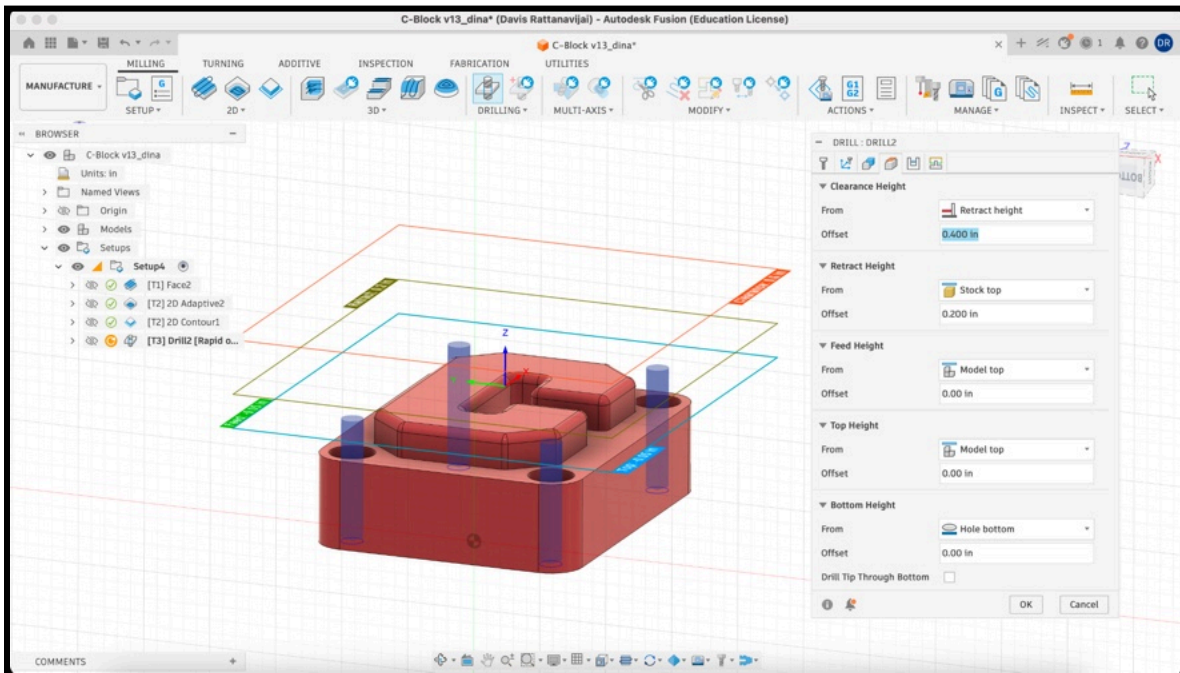
1. Select the drilling event
2. Select your drill tool and verify the feed and speeds



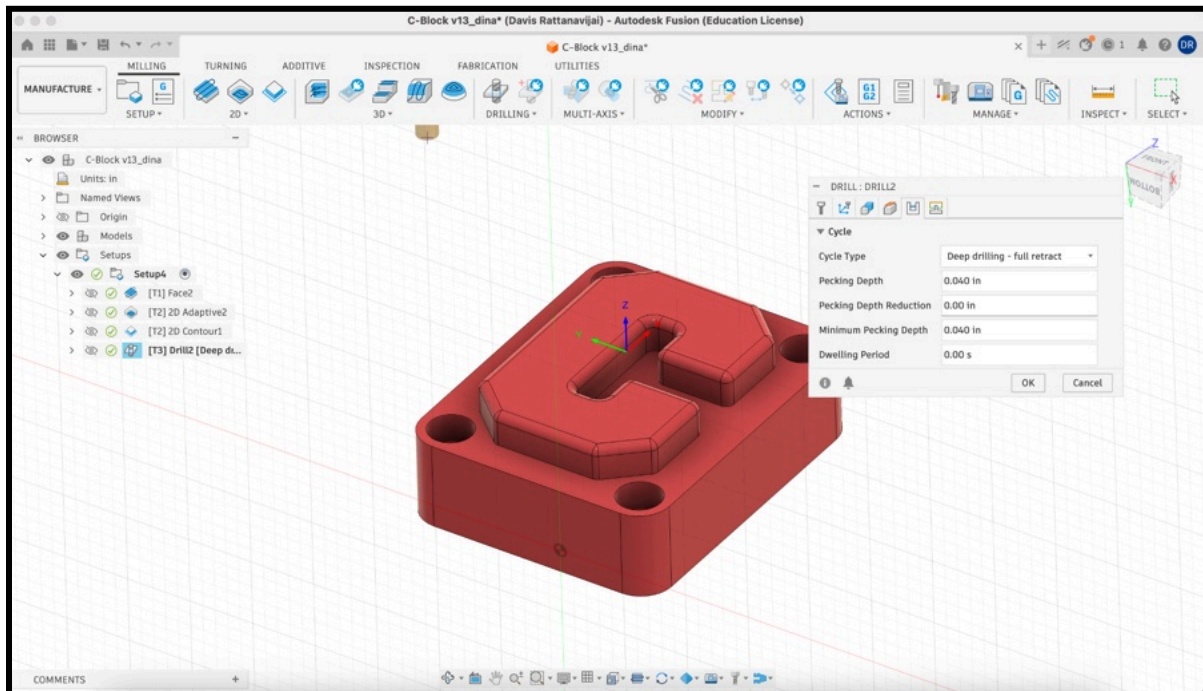
3. For the Geometry tab, select the holes you want. You can select them one by one, or, since they are the same size, check Select Same Diameter to select them all at once.



4. Then go to the Heights tab and make sure all the heights are reasonable. These are the heights the toolpaths follow, so it is important to set them correctly.



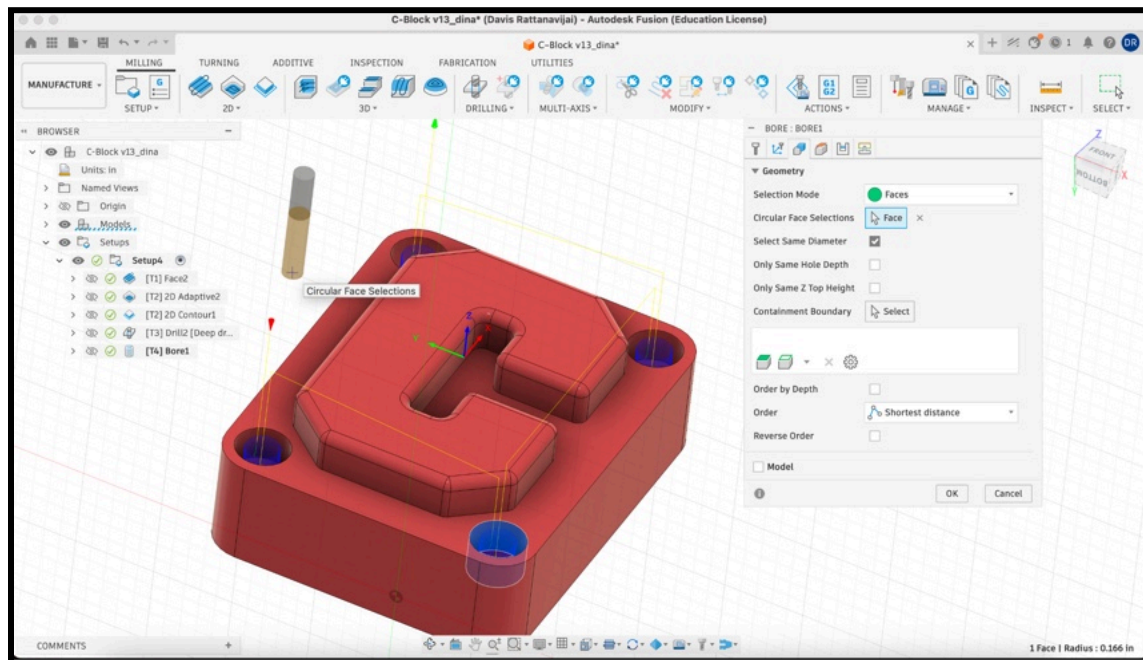
5. For the Cycle, select Deep Drilling – Full Retract. This is the equivalent of peck drilling.



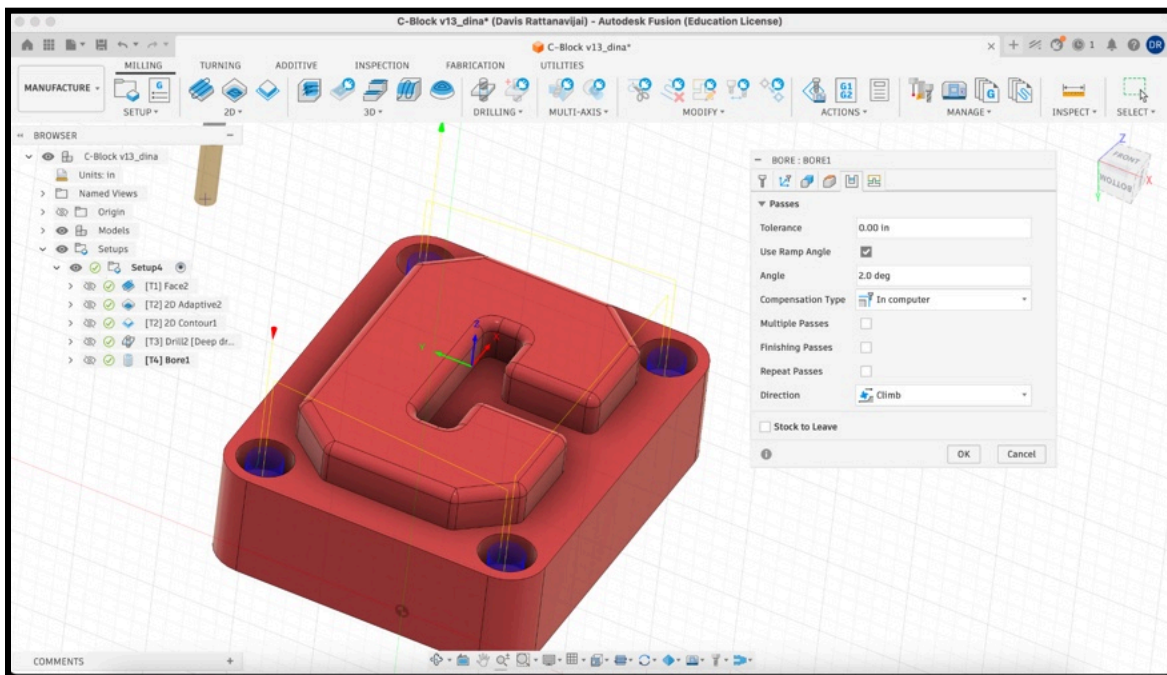
6. Press okay and view the tool path and verify.

Part 7: Boring the Holes

1. Select the bore event
2. Select your 1/8 end mill and verify that the feeds and speeds are correct



3. For the Passes, enter the values shown below. The angle should be relatively small, around 2 degrees, as this controls the angle at which the end mill bores downward.

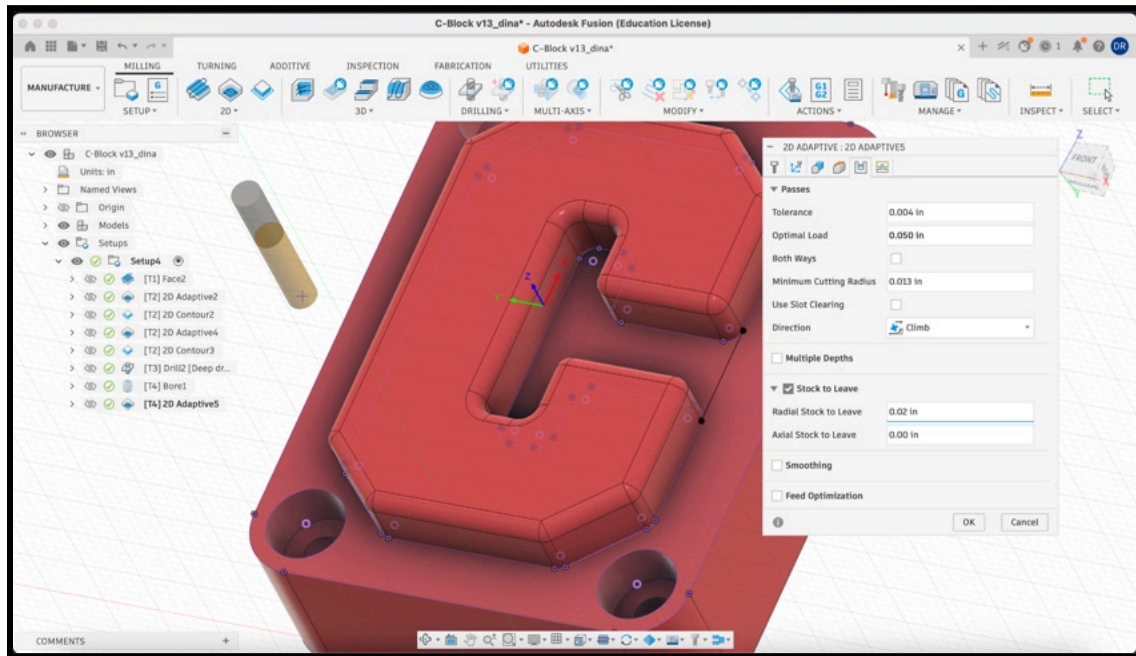


4. Click OK and check the toolpath. Make sure the helix is reasonable for the end mill, and verify the retracts and tool movements so the tool does not crash into the stock.

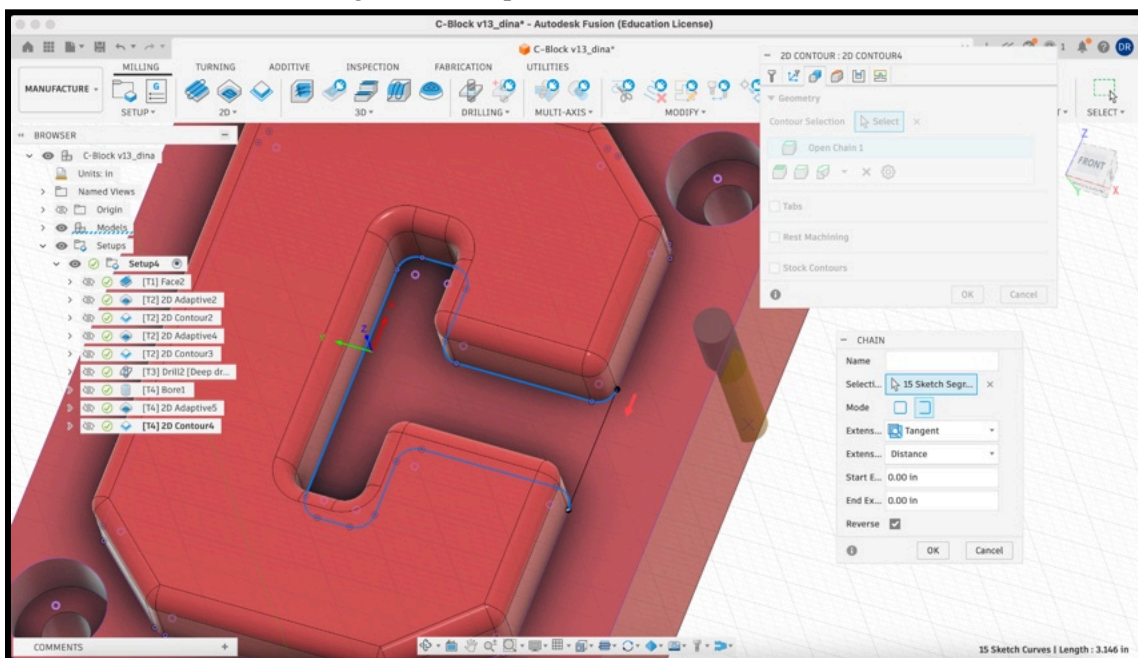
Part 8: Milling the C Inner Outline

In this part we will mill out the general inner profile of the C.

1. Select the Adaptive event and keep the 1/8-inch end mill. We are going to use the adaptive + contour combination again, so set Axial Stock to Leave to 0 and Radial Stock to Leave to 0.02. Then select the outline of the inner part of the C. You should have also created a sketch at the beginning that this will reference.



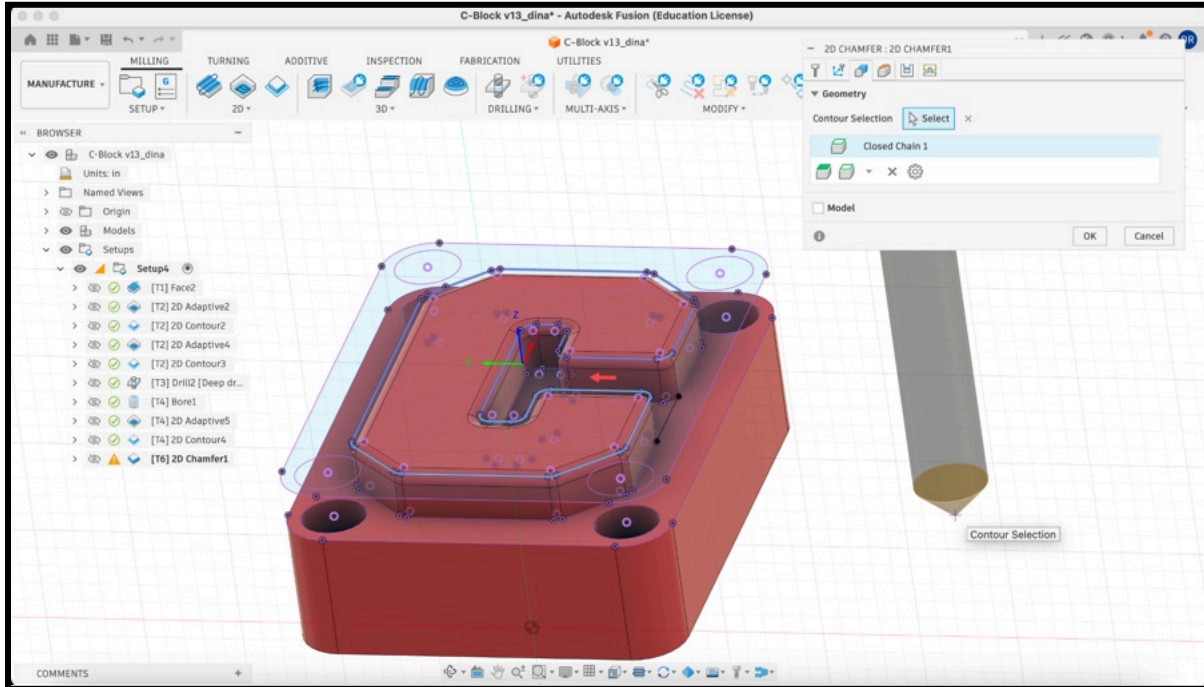
2. Do the contour event using a derived operation.



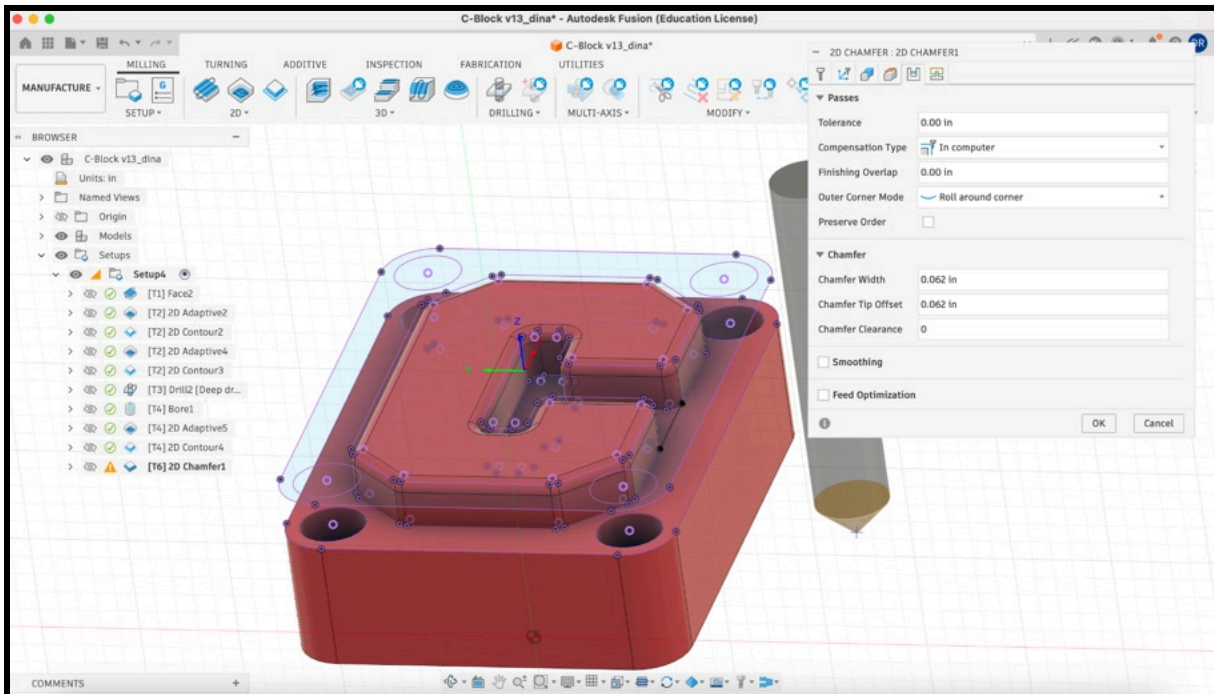
3. Press okay to save and verify the tool path.

Part 9: Chamfering

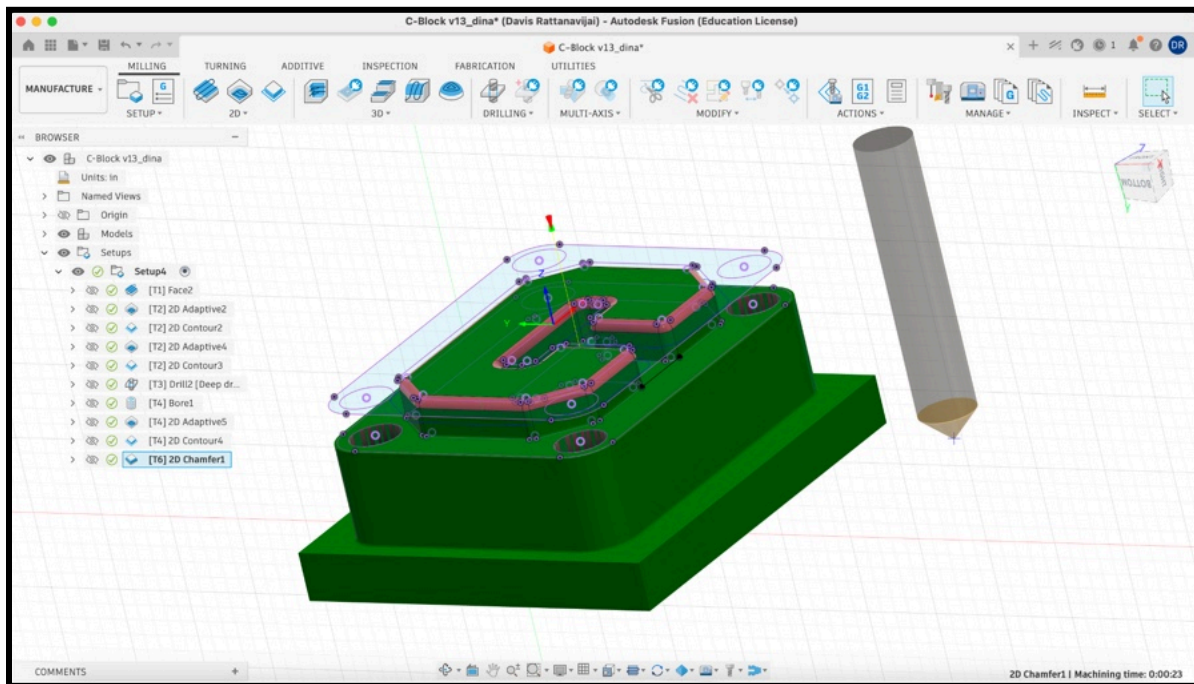
1. Select the chamfering event and select your chamfer tool
2. For Geometry, use a Closed Chain and select the outer outline of the C-Block from the sketch you made at the start.



3. For the Passes tab, set the Chamfer Width to 0.062 and the Tip Offset to 0.062



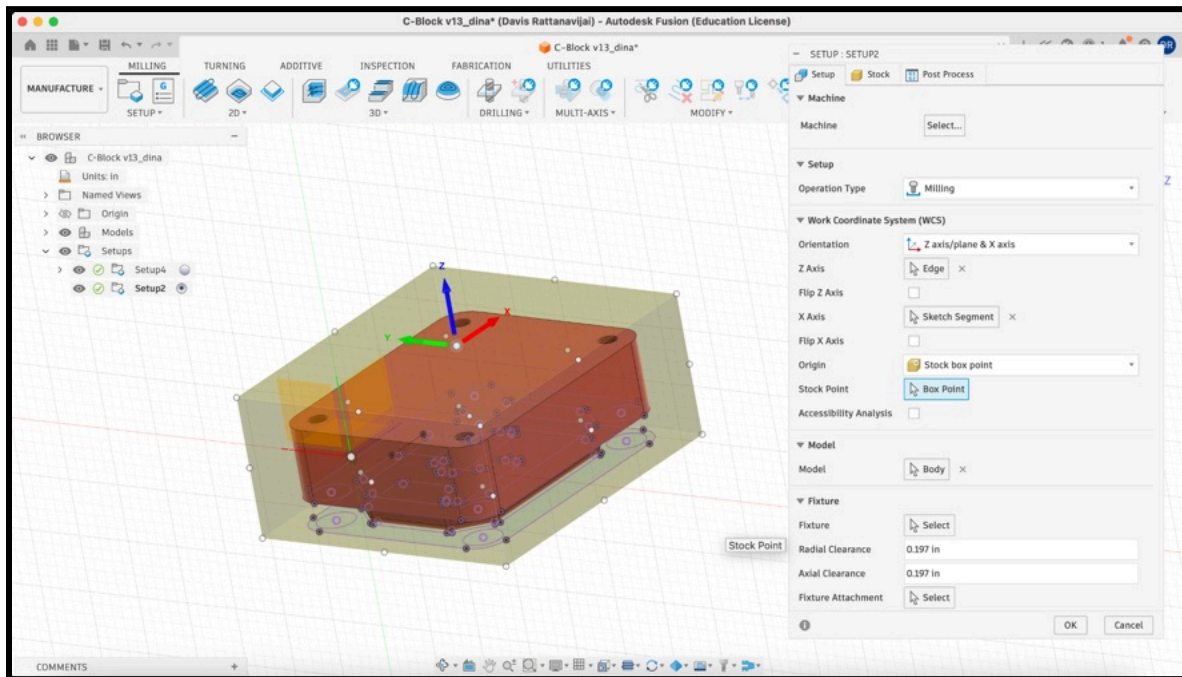
4. Press okay and double verify the chamfers look normal. If you see pink that is normal.



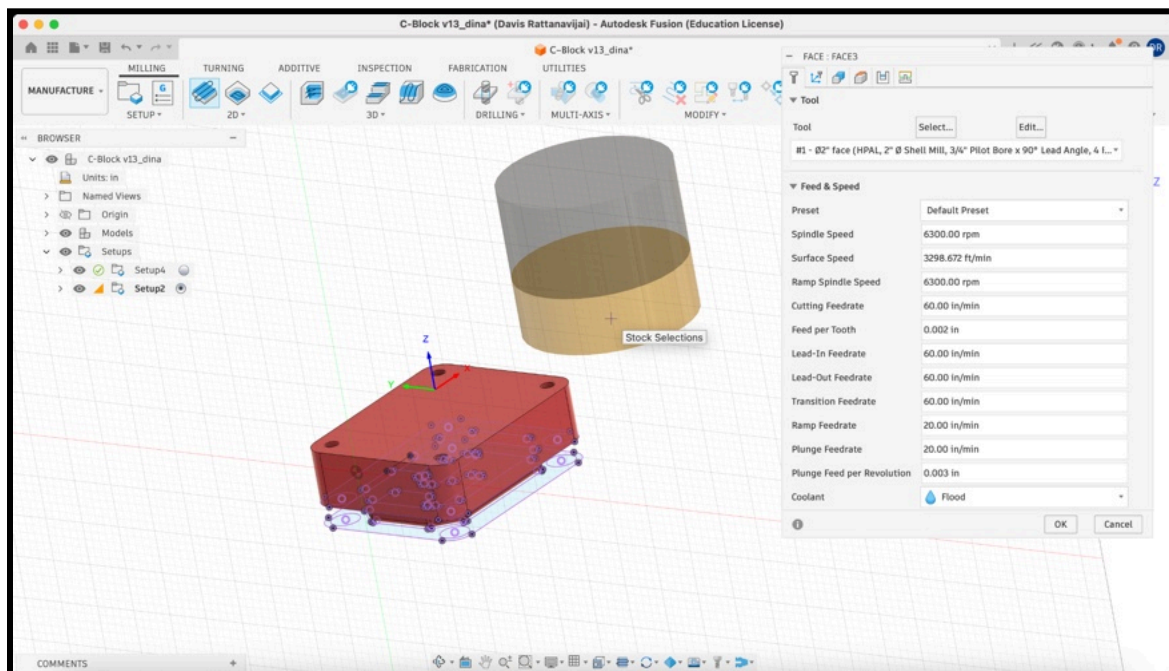
Part 10: Milling the Leftover Stock

This is the final step. Before running the program, change how the piece is positioned in the vise. Since we are moving the stock, we must create a new setup in Fusion.

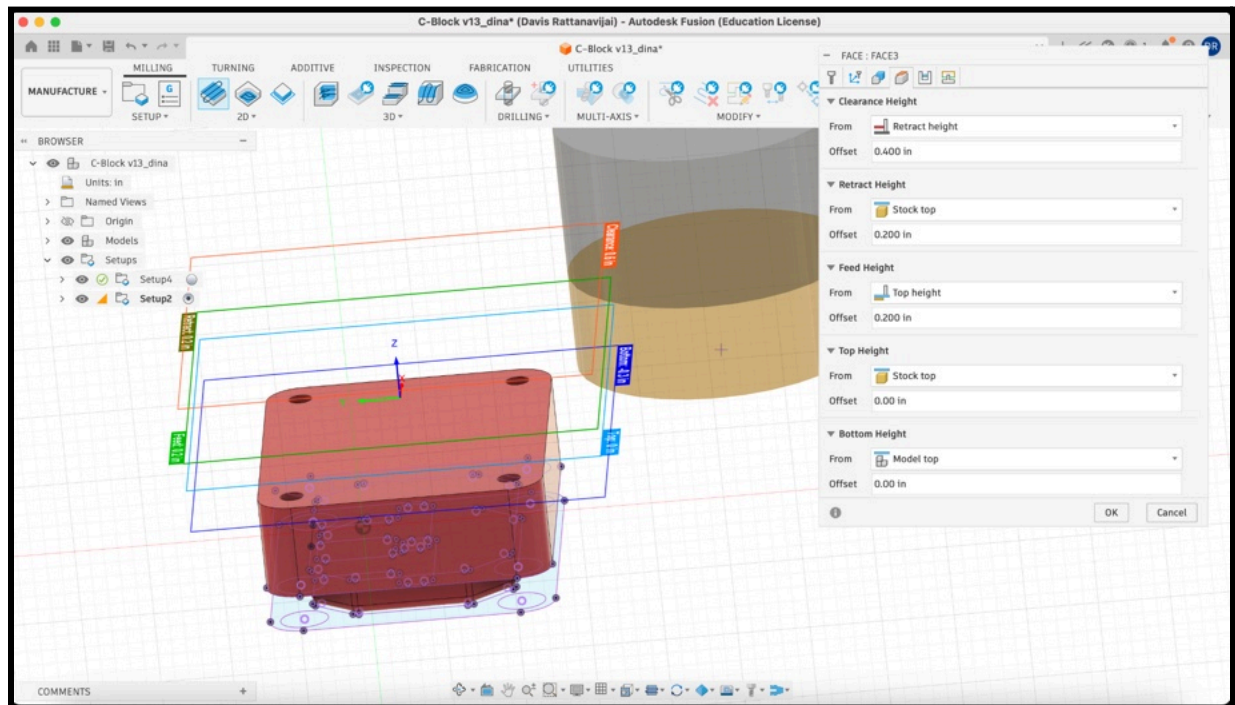
1. Create a new set up from the set up tab in the top left corner
2. Make sure your coordinate system looks like the one below. Importantly, Z points up, X runs along the long direction, and the coordinate system is centered.



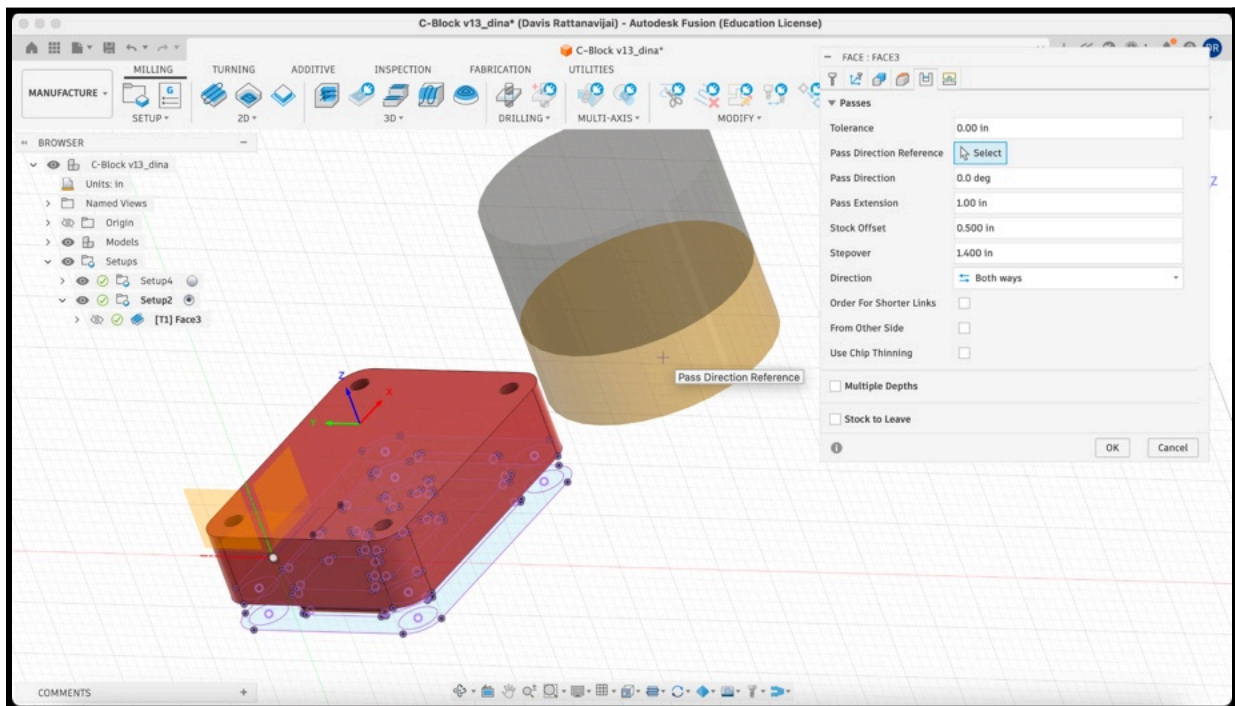
3. Now select a facing event and use your shell mill. Again, double check feeds and speed



4. Check the heights. Make sure the Bottom Height is set to the bottom of the C-Block, which determines how much material we want to cut.



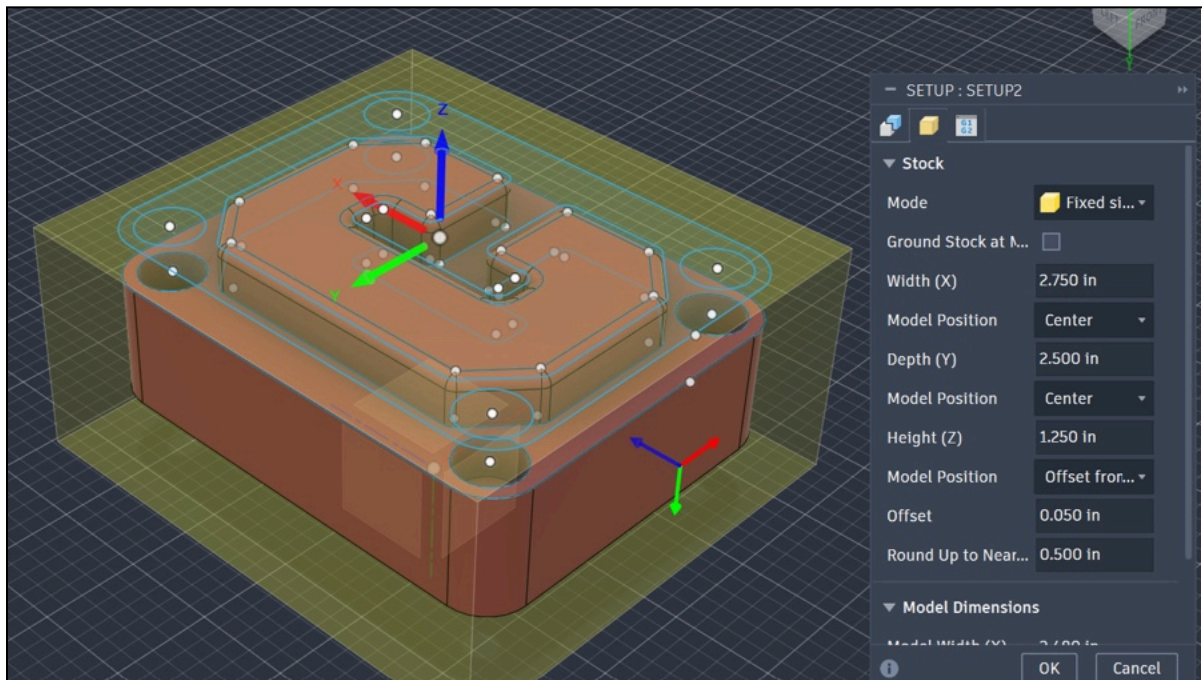
5. For the Passes tab, select the correct pass direction and stock offset. Again, the pass direction should be toward the vise to minimize stock slippage.



6. Press okay, and Verify what the program will face the whole surface (same as part 3).

Part 11: Running Your Program on the HAAS

1. Power on CNC machine, release E-Stop, press power up
2. Get a stock piece of aluminum and measure its dimensions. These dimensions may require you to alter your stock setup in the C Block CAM. Once you adjust this, there will be an error in the program. To fix this, go to the Browser menu, highlight the Setup tab, and press **Ctrl + G** to update your CAM on your computer.



Example: The stock used in this example was measured to be 2.75" x 2.50" x 1.25"

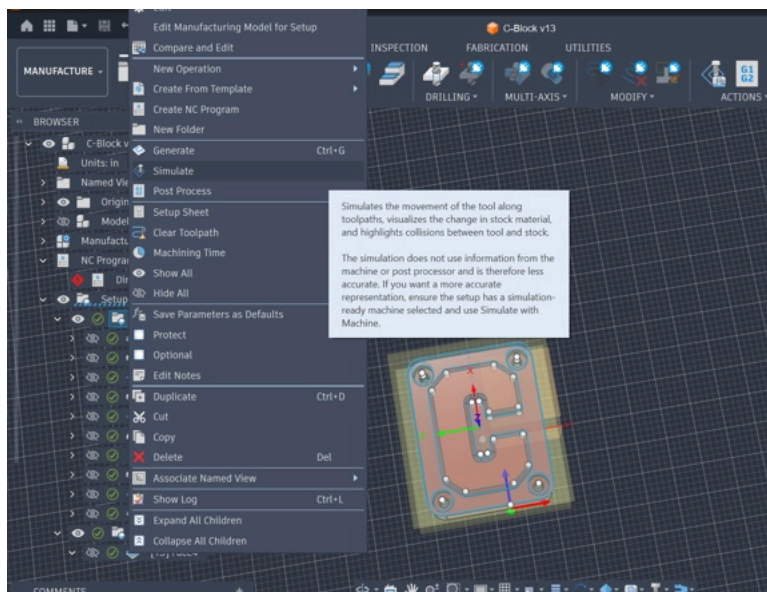
3. Upload your program to the machine using a USB. Copy your code to Memory through List Programs using **F2**. Press the **MEMORY** button and select your program from the list. You should see it displayed on the screen.
4. **EDIT -> probing -> tool offsets**, probe all the tools you will be using and make sure the tool number matches the order of your tools in your C Block CAM tool table.
Remember to switch a tool use **MDI-> type in T # tool -> ATC FWD or ATC REV(will just be the direction the carousel spins)**.

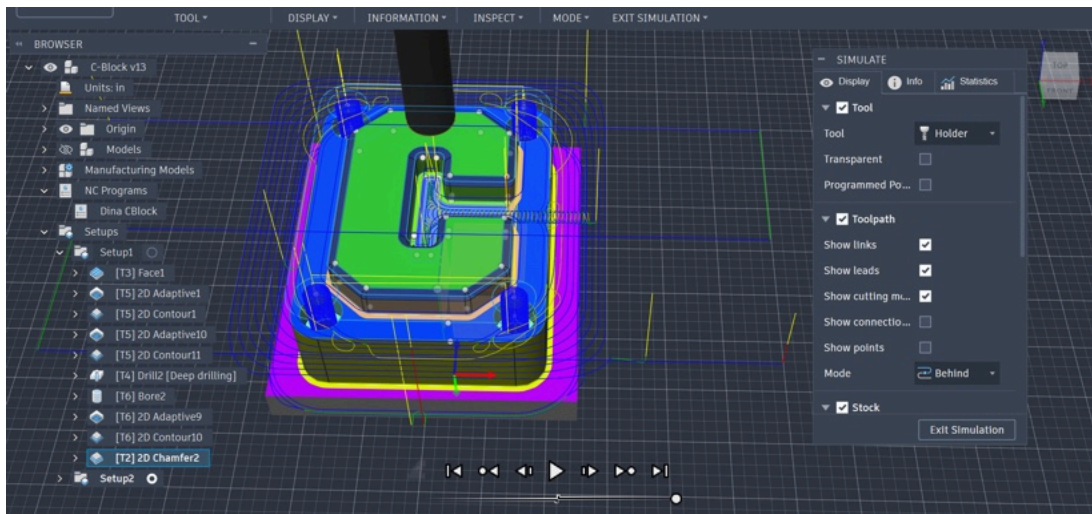
Once done with probing the tool you are using be sure to check this with the height of Z min to make sure the tool doesn't hit the vice. This is important to ensure nothing will crash into the vice and damage the CNC. To do this read off the Z min in your code. And manually bring your tool to that Z minimum position offset value reading off of the screen using handle jog. Do this off of the vice so you can see if the tool will hit. Repeat this process for tools that are longer. Adjust Z min as needed.

	Name ^	Corner radius	Diameter	Flute length
C-Block v13				
	1 - Ø1/4" 45° (1/4" Dia. 45° Carbide Chamfer Mill, 4 Flute, TiAlN Coated, 1/8" LOC ...	0 in	0.25 in	0.125 in
	2 - Ø3/8" 45° (3/8" Dia. 45° Carbide Chamfer Mill, 2 Flute, TiCN Coated, 3/16" LOC...	0 in	0.375 in	0.187 in
	3 - Ø2" (HPAL, 2" Ø Shell Mill, 3/4" Pilot Bore x 90° Lead Angle, 4 Flute, Positive/...	0 in	2 in	0.67 in
	4 - Ø0.159" 135° (#21 Ø Cobalt Jobber Drill, 135° Point Angle x 30° Helix, 2 Flute, ...	0 in	0.159 in	2.125 in
	5 - Ø1/2" R0.015" (1/2" Ø Carbide End Mill, 3 Flute, Uncoated, 1/2" Shank x 1-1/4" ...	0.015 in	0.5 in	1.25 in
	6 - Ø1/8" (1/8" Ø Carbide End Mill, 3 Flute, Uncoated, 1/8" Shank x 7/16" LOC, Squ...	0 in	0.125 in	0.438 in
Tool Holders				

Example: Tool 1 in the ATC of the CNC you are using should be the ¼ chamfer end mill in this case. Tool 2 should be ⅜ chamfer mill.... The order doesn't have to match this table but should match your table.

- Now switch the tool to the spindle probe to probe the stock. Probe the center of the block using EDIT → Probing → Work Offsets → Rectangular Block.
- After you are done probing, switch your tool before moving to step 6. The coolant should never be on when the spindle probe is out. Since it is expensive we don't want to damage it in any way. Now that the stock and all the tools are set, use a refractometer(see CNC overview video) to ensure coolant levels are correct. Check your coolant nozzle as well. The CLNT UP and CLNT DOWN in the control panel on the left adjusts the nozzle angle. You want to make sure it is above the tool, so that the coolant nozzle doesn't hit the vice. This is important so that nothing crashes and breaks. In the MDI button tab if you hit coolant at any time while running your program you can toggle the coolant on and off.
- Now check your work in **MEMORY** → **GRAPHICS** and also check the simulation for your **C Block CAM Setup 1**. Everything should look the way you want it to. Make sure to have contours after your adaptive clearings and have **0 for all your axial stock to leave**. You will see stock left on the part if this is not set to 0.





8. Now you are ready to start your program. Start on **Single Block** for the first few cuts, and then toggle it off once you are comfortable with the program. Lower your rapids as well. The coolant should be on the whole time. Press **Cycle Start** when you are ready, and always have your hands hovering near **Feed Hold**. If something is wrong, press Feed Hold immediately.
9. After you are done with Setup 1, upload the program for Setup 2 and flip the block over. Now we will cut off the lip at the bottom of our C Block.

When running this setup, be aware that slipping is very likely, so you may want to clean off the vice and wipe the piece with a towel before running the program. Also, make sure your passes are in the **Y direction** and not the X direction to reduce slipping.

After running this setup, you will have your completed C Block. YAY!